Deep Learning with Neural Networks

Unsupervised Deep Learning (I): autoencoders

Alejandro Lancho Serrano, alancho@ing.uc3m.es

Material original por:
Pablo Martínez Olmos,
José Carlos Aradillas,
Ignacio Peis

Unsupervised vs Supervised

Supervised learning

Data: ***(x, y)***
x is data, ***y*** is label

Goal: Learn function to map
x → ***y***

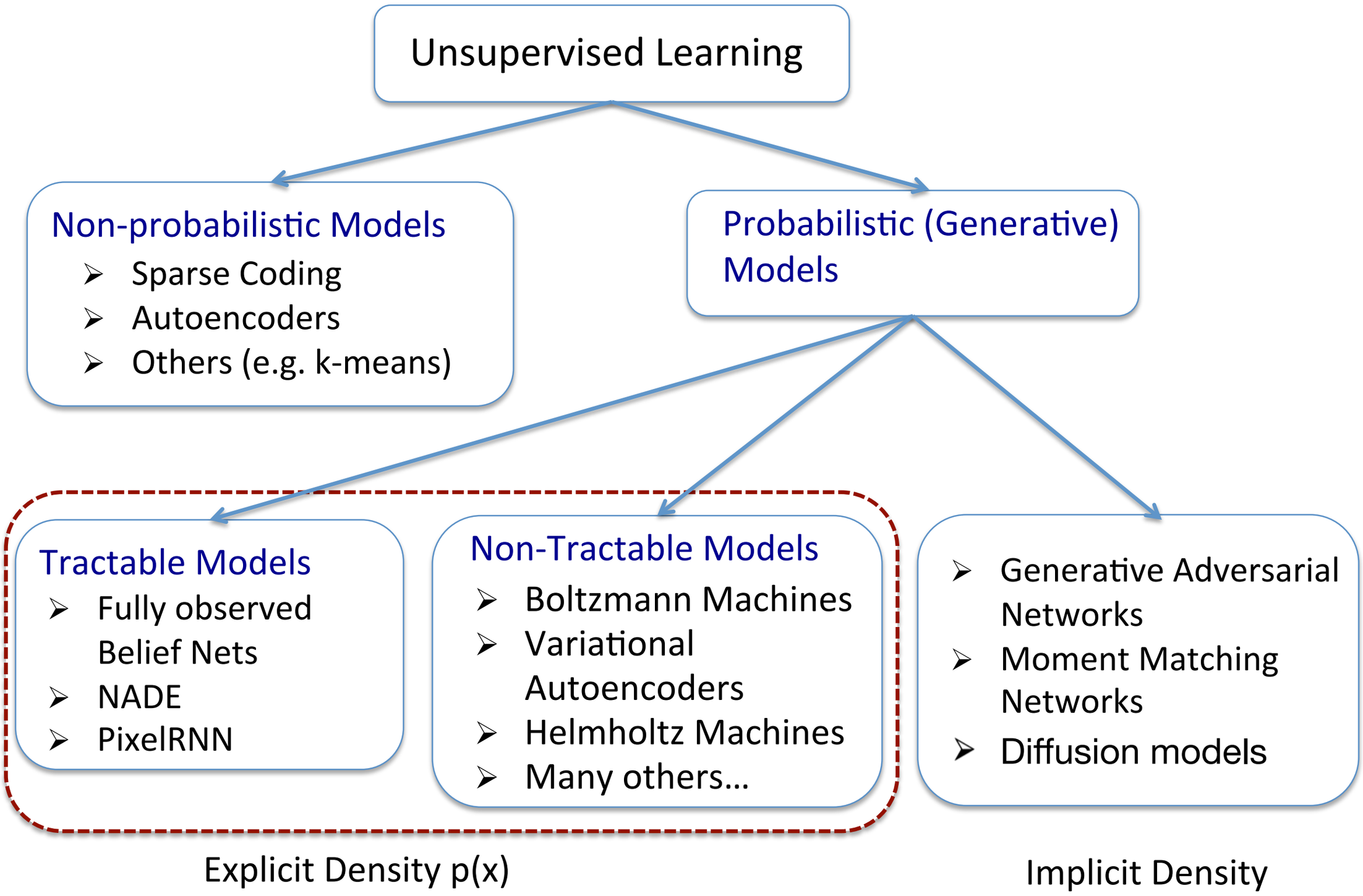
Examples: Classification, regression, object detection, semantic segmentation, etc.

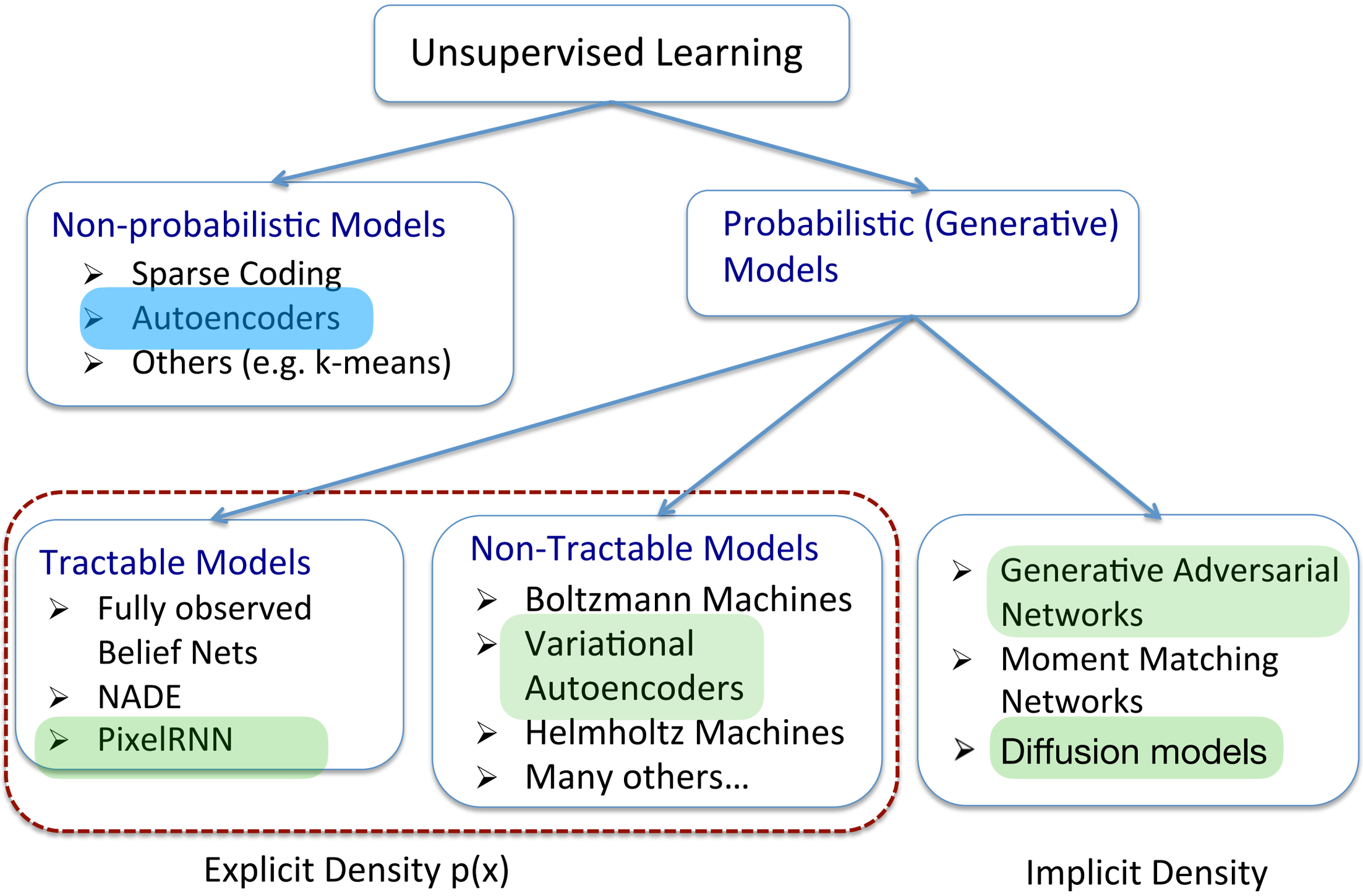
Unsupervised learning

Data: ***x***
x is data, no labels!

Goal: Learn the *hidden* or *underlying structure* of the data

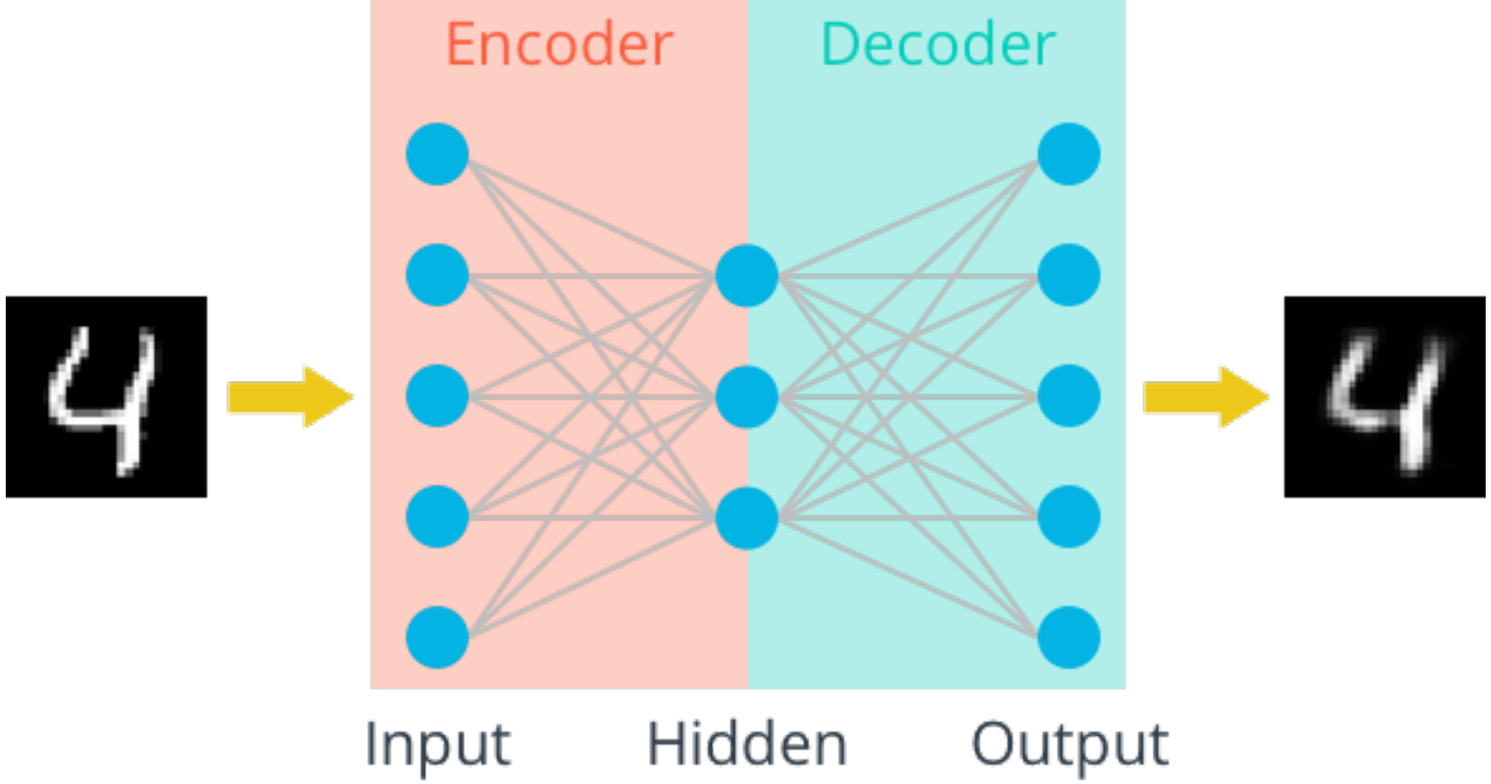
Examples: Clustering, feature or dimensionality reduction, etc.


Unsupervised Learning
Non-probabilistic Models
Sparse Coding
Autoencoders
Others (e.g. k-means)
Probabilistic (Generative) Models
Tractable Models
Fully observed Belief Nets
NADE
PixelRNN
Non-Tractable Models
Boltzmann Machines
Variational Autoencoders
Helmholtz Machines
Many others…
Generative Adversarial Networks
Moment Matching Networks
Diffusion models
Explicit Density p(x)
Implicit Density


Unsupervised Learning
Non-probabilistic Models
Sparse Coding
Autoencoders
Others (e.g. k-means)
Probabilistic (Generative) Models
Tractable Models
Fully observed Belief Nets
NADE
PixelRNN
Non-Tractable Models
Boltzmann Machines
Variational Autoencoders
Helmholtz Machines
Many others…
Generative Adversarial Networks
Moment Matching Networks
Diffusion models
Explicit Density p(x)
Implicit Density

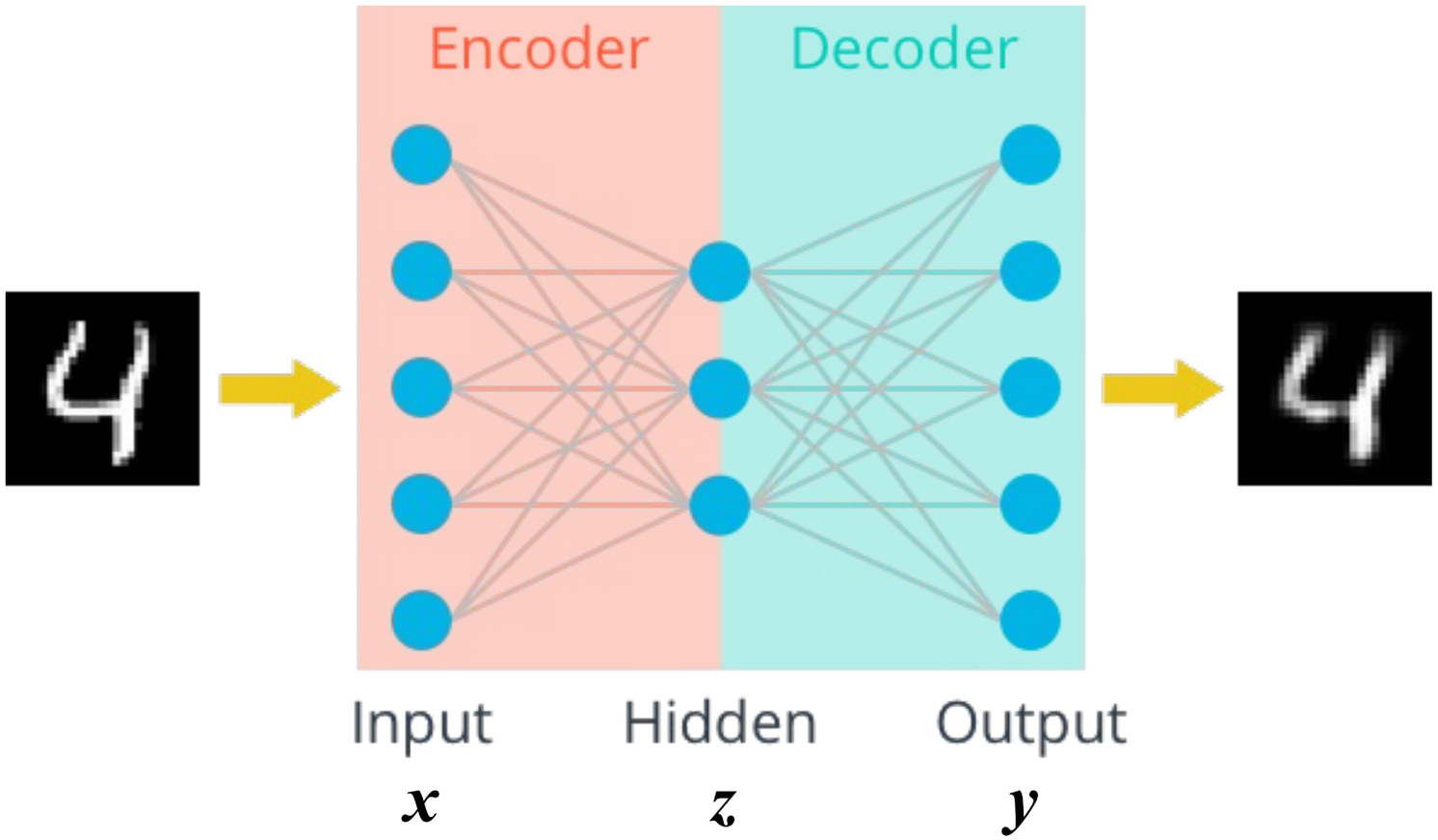
Deep Autoencoder

- An autoencoder is a neural network that is trained to attempt to copy its input to its output
- Internally, it has a low-dimensional hidden representation
- Two parts: encoder and decoder
- Autoencoders have been traditionally used for dimensionality reduction or feature learning



$$L(\boldsymbol{x}, g(f(\boldsymbol{x})))$$

Deep Autoencoder



- The “encoder” learns mapping from the data, x, to a low-dimensional latent space, z

$$\boldsymbol{z} = f(\boldsymbol{x})$$

- The “decoder” learns mapping back from latent space, z, to a reconstructed observation, y

$$\boldsymbol{y} = g(\boldsymbol{z}) = g(f(\boldsymbol{x}))$$

- The loss function doesn’t use any labels

$$L(\boldsymbol{x}, g(f(\boldsymbol{x})))$$

Dimensionality of latent space → reconstruction quality

Autoencoding is a form a compression

2D latent space

5D latent space

Ground Truth

© MIT 6.S191: Introduction to Deep Learning
introtodeeplearning.com

Autoencoders for representation learning

Bottleneck hidden layer forces network to learn a compressed latent representation

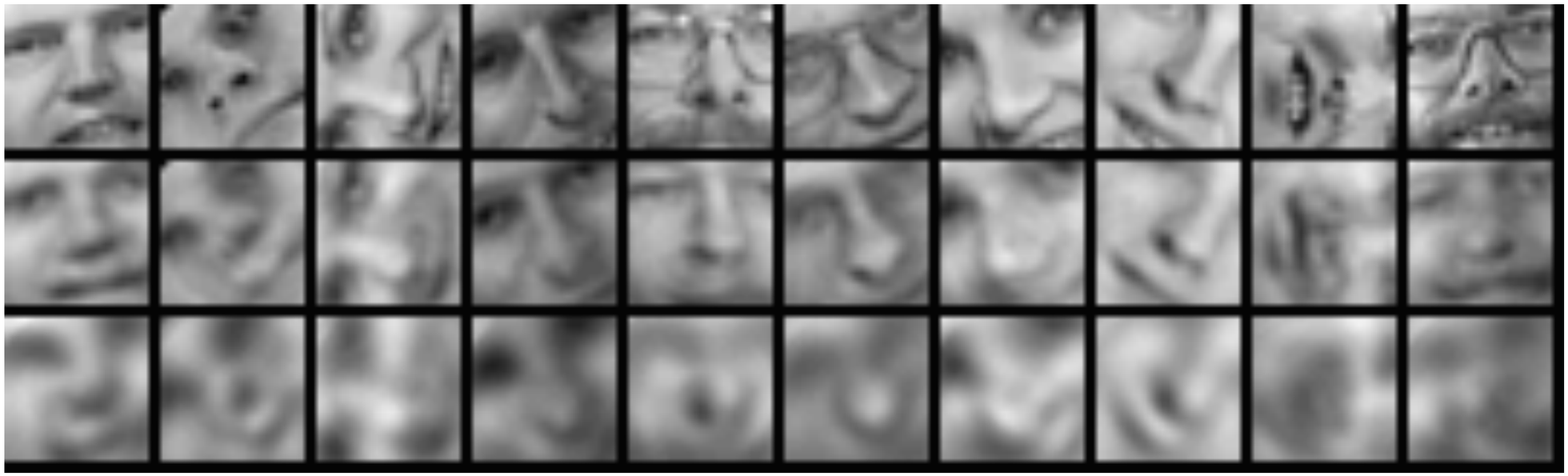
Reconstruction loss forces the latent representation to capture (or encode) as much "information" about the data as possible

Autoencoding = **Auto**matically **encoding** data; "Auto" = **self**-encoding

© MIT 6.S191: Introduction to Deep Learning
introtodeeplearning.com

Deep Autoencoder

• 25x25 – 2000 – 1000 – 500 – 30 autoencoder to extract 30-D real-valued codes for Olivetti face patches.

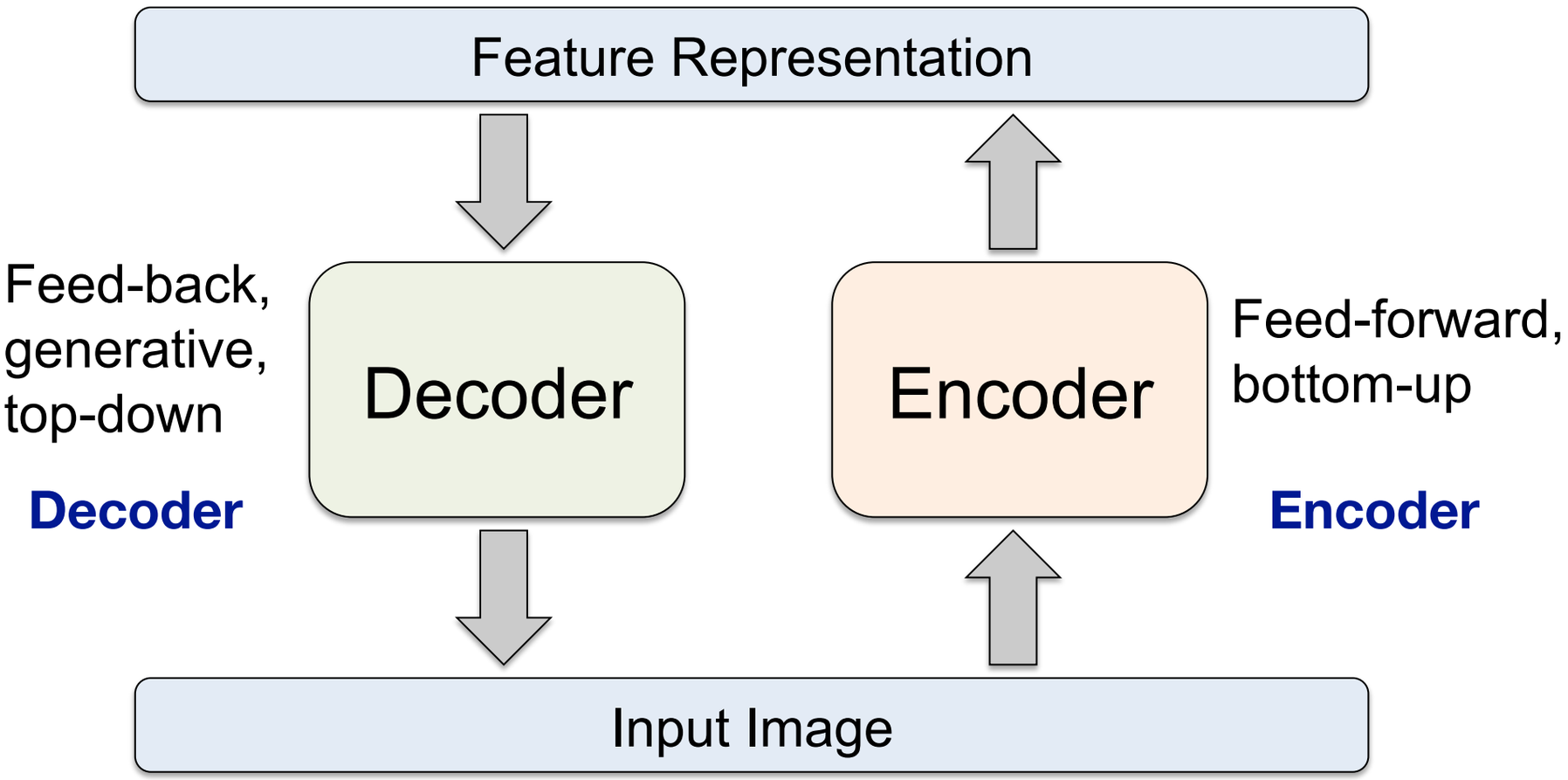


• **Top**: Random samples from the test dataset.

• **Middle**: Reconstructions by the 30-dimensional deep autoencoder.

• **Bottom**: Reconstructions by the 30-dimentinoal PCA.

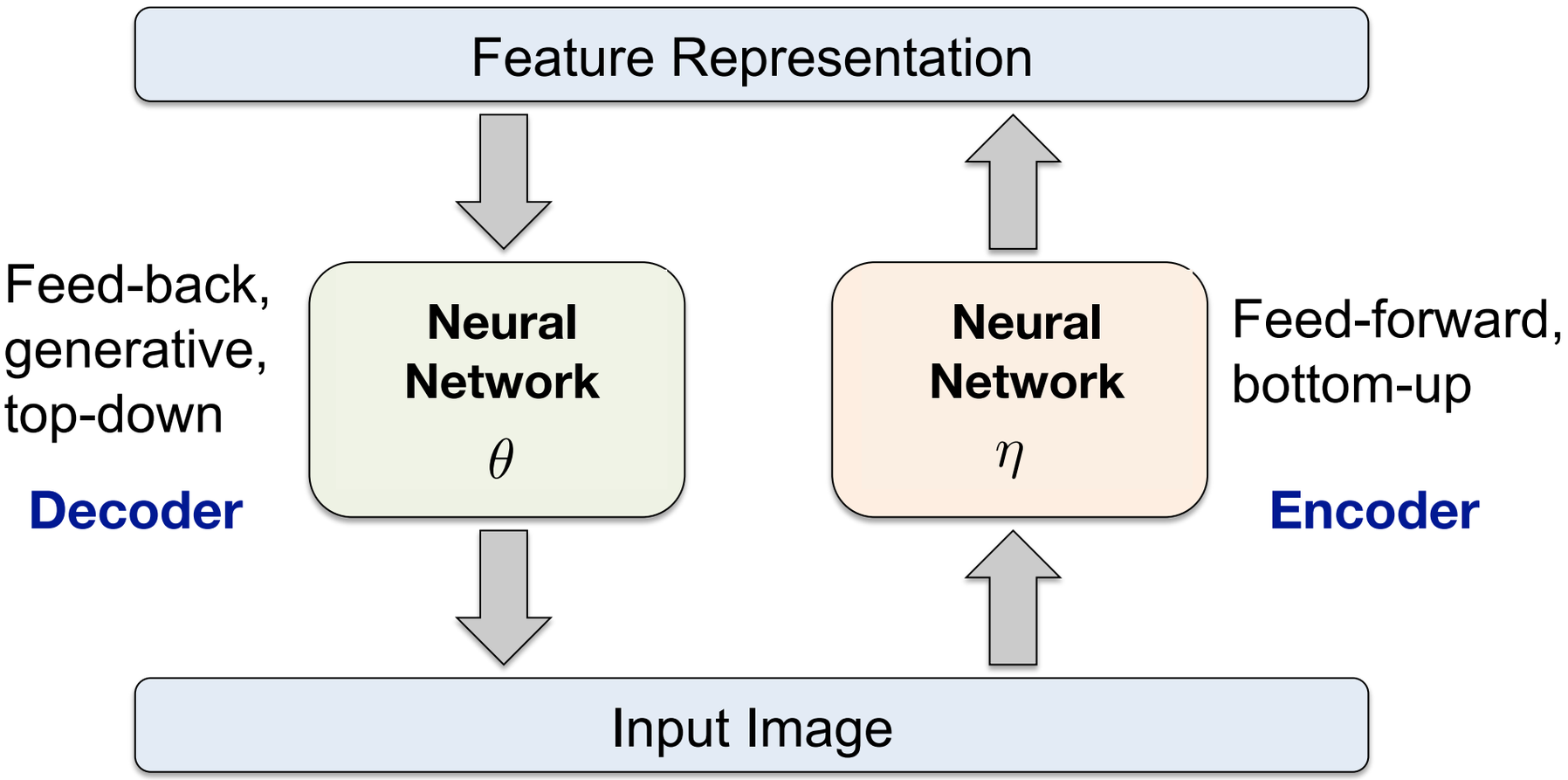
Deep Autoencoders



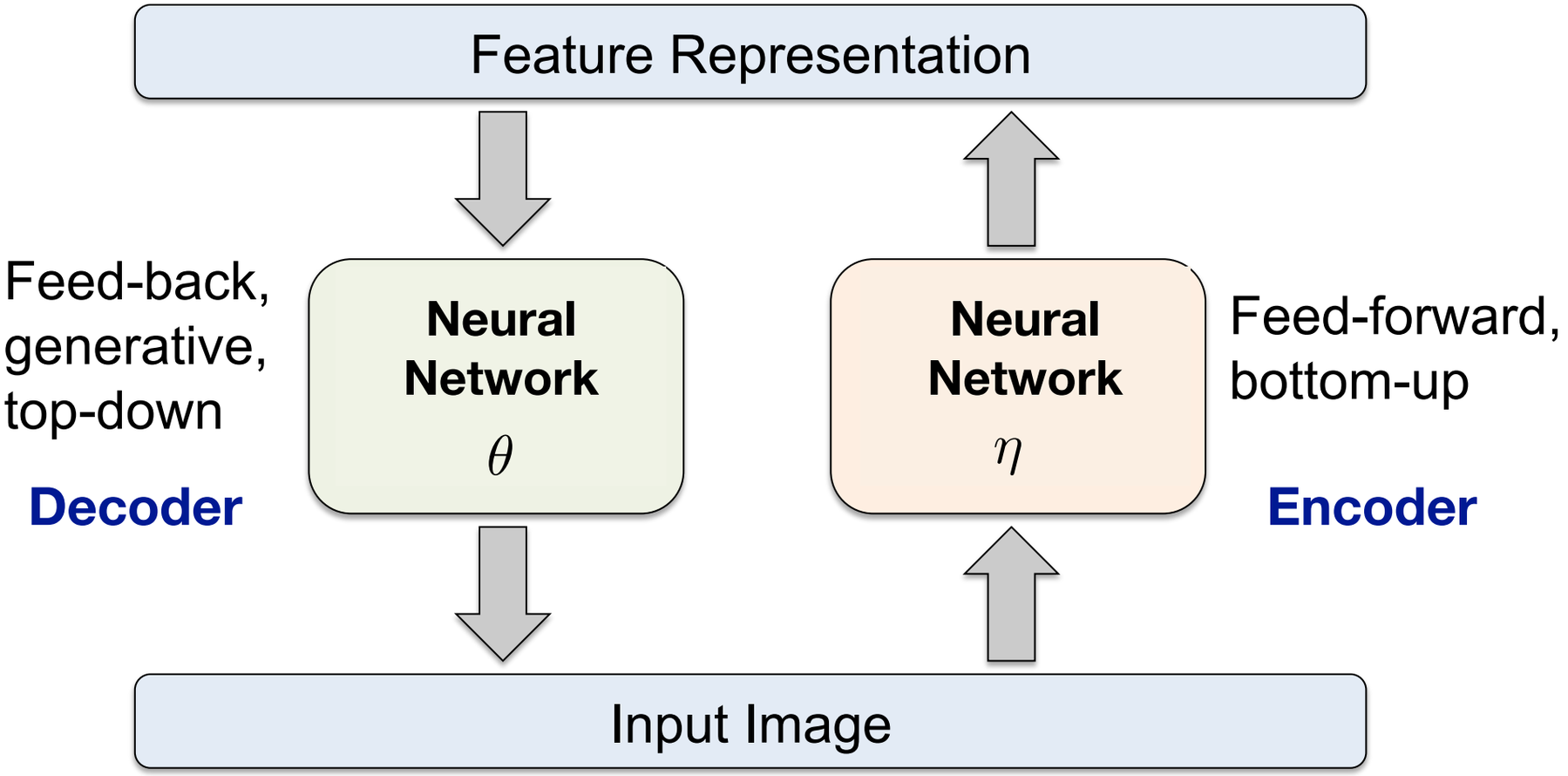
Decoder

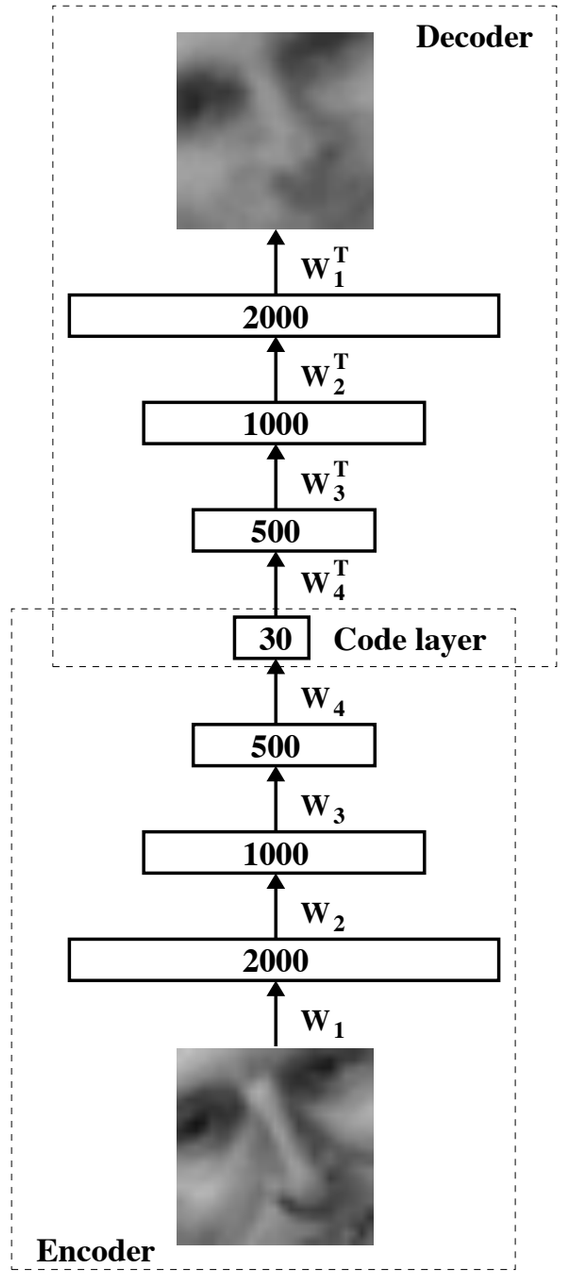
Encoder

Deep Autoencoders

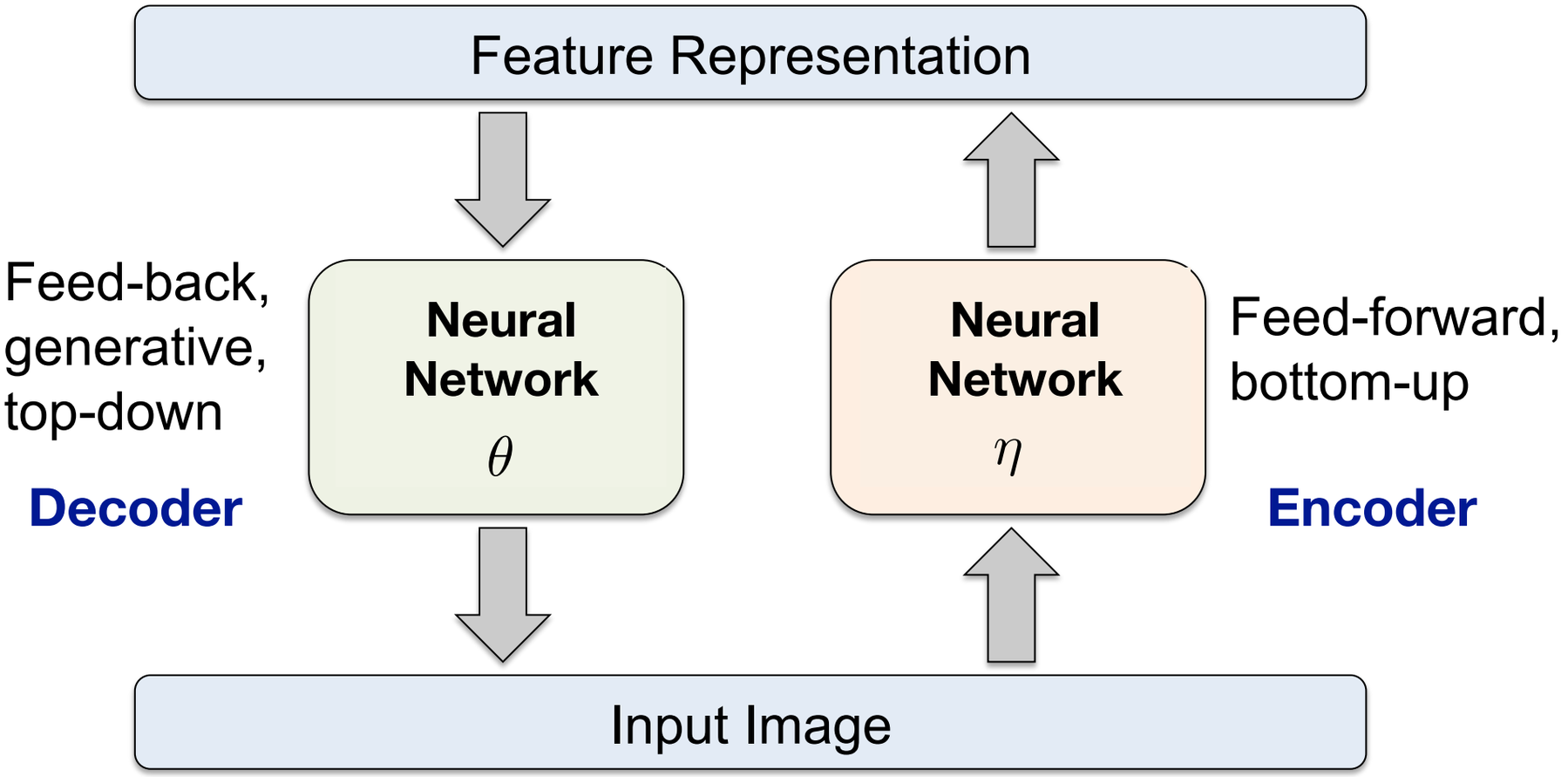


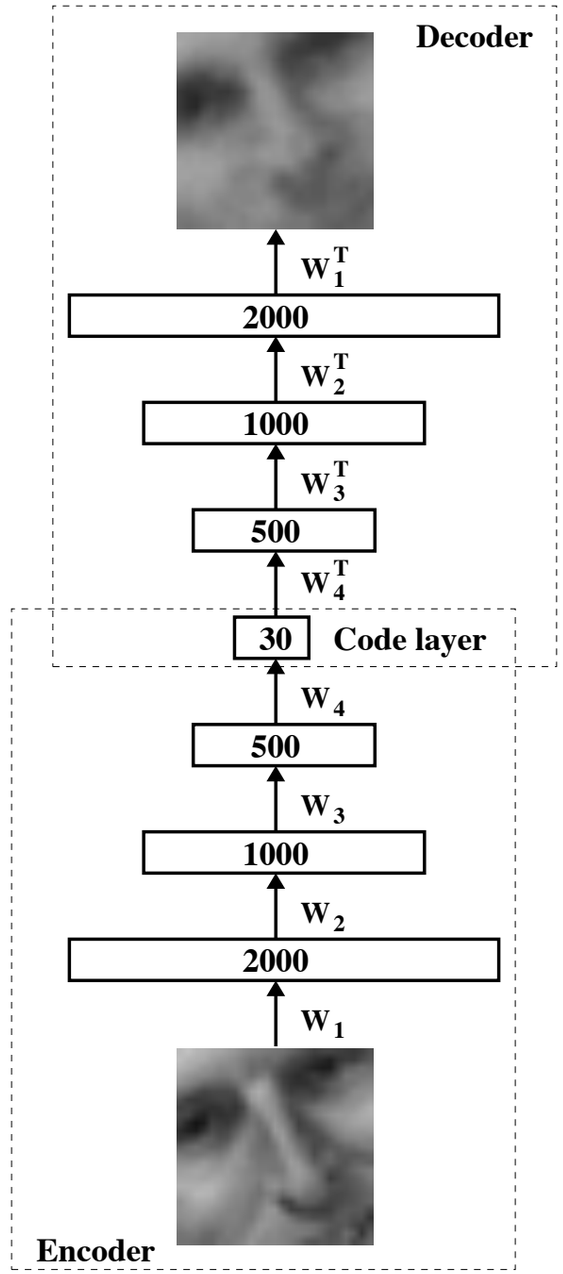
Deep Autoencoders





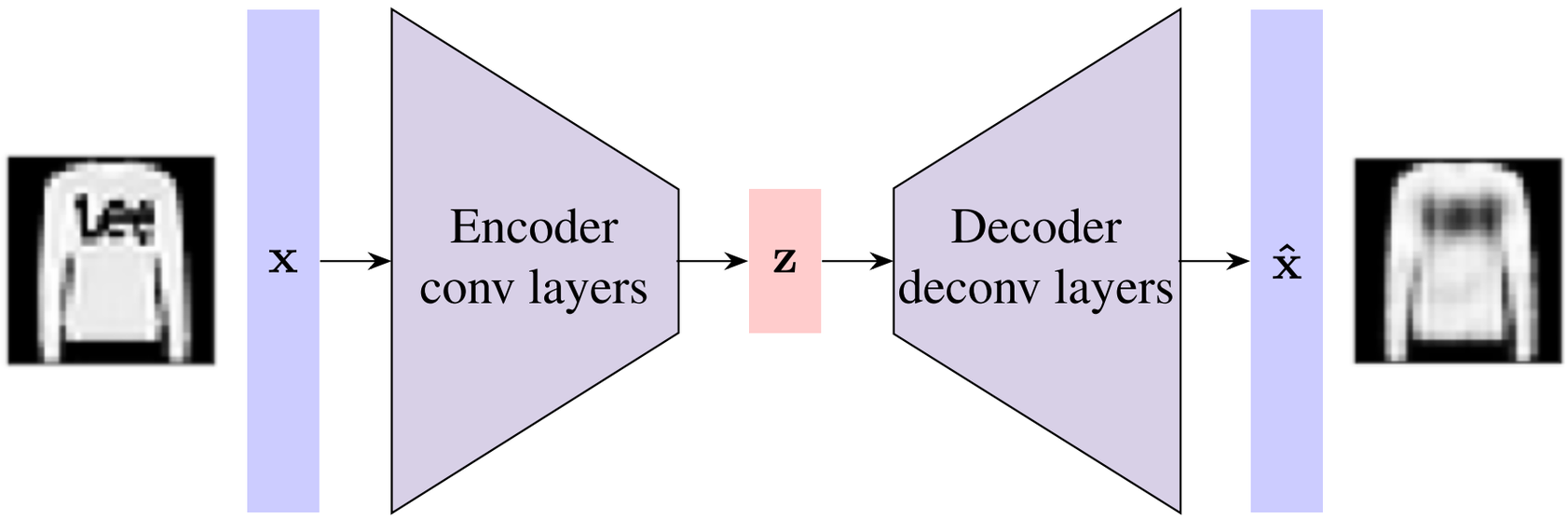
Deep Autoencoders





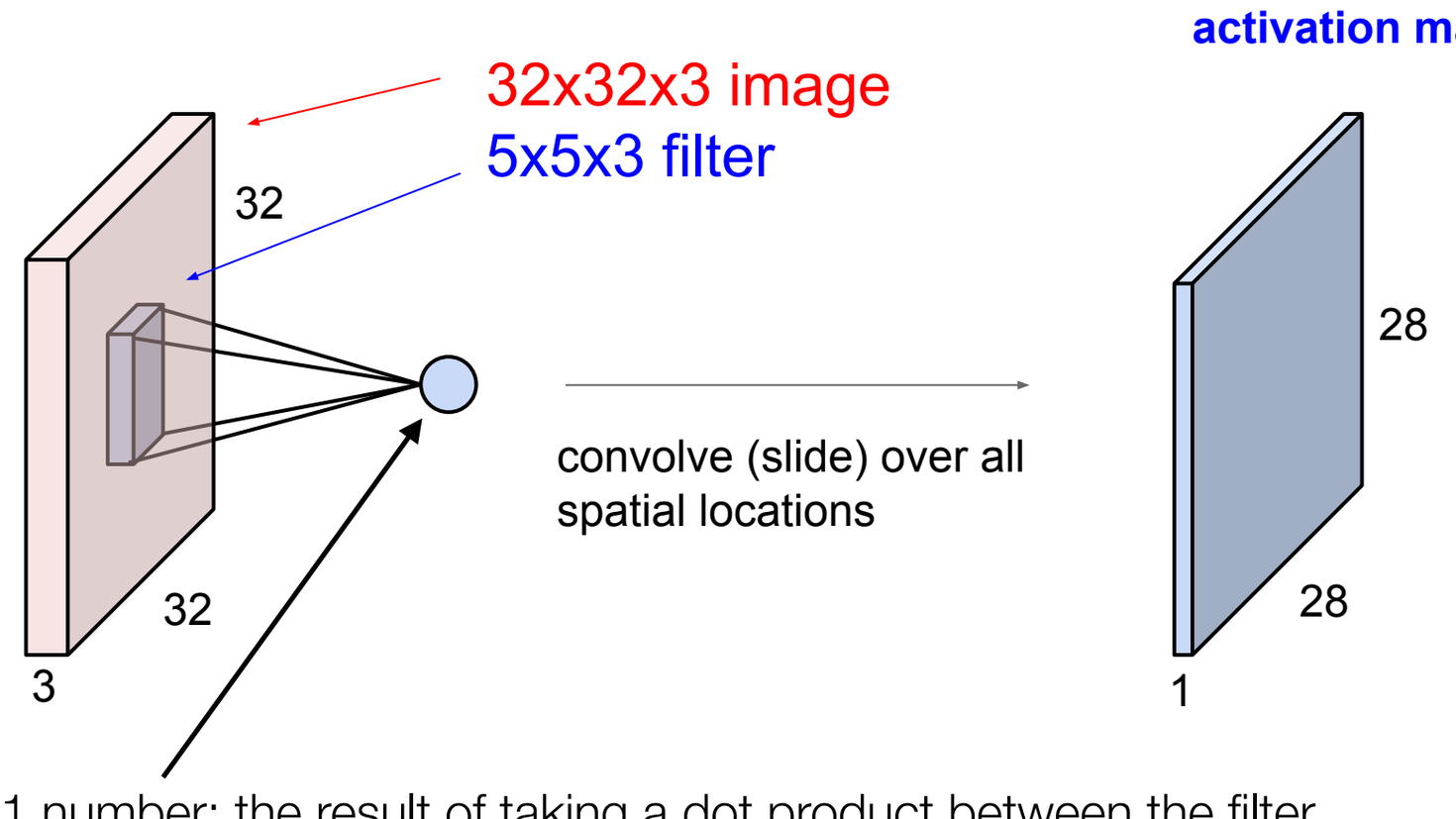
$$\mathcal{L}(\eta, \theta) = \frac{1}{N} \sum_{n=1}^{N} \mathcal{L}_n \left(\mathbf{x}_n - D_\theta \left(E_\eta(\mathbf{x}_n) \right) \right)$$

Convolutional Autoencoders



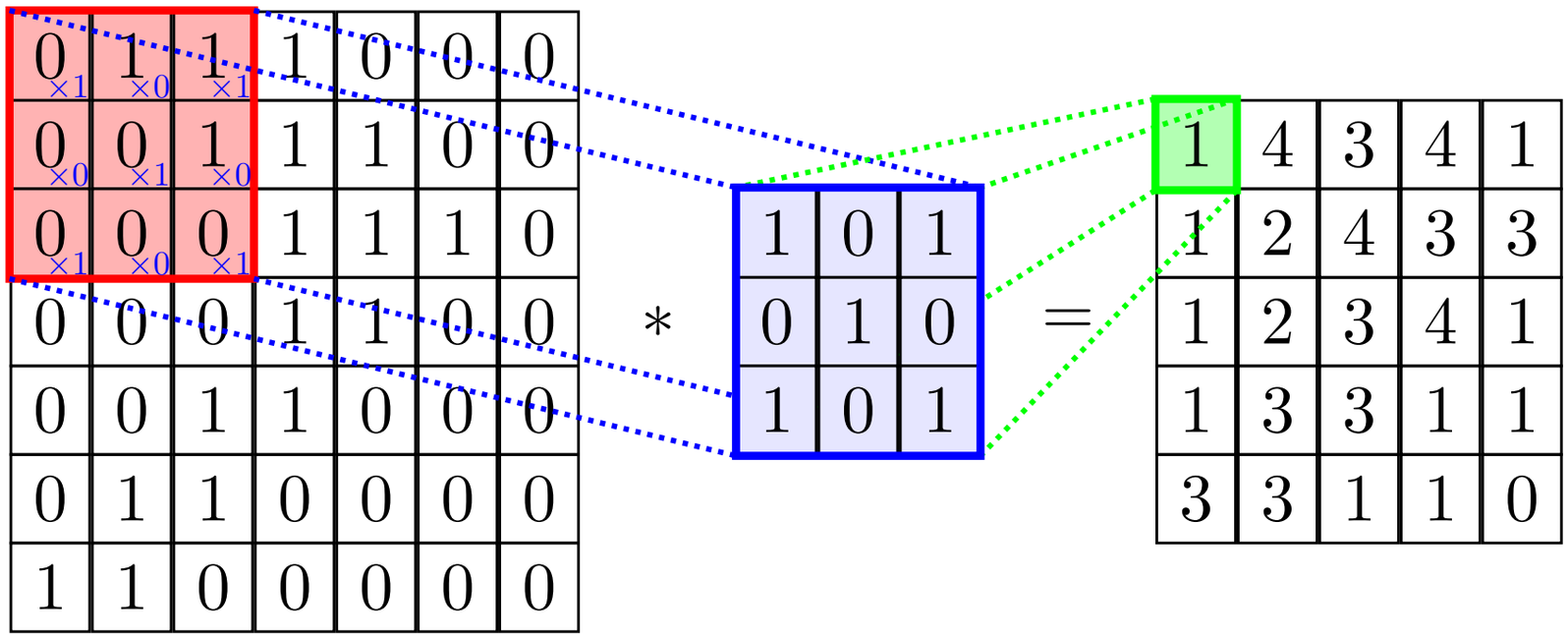
- Considered convolutional autoencoder if one or more (de)-convolutional layers

Convolution layers



1 number: the result of taking a dot product between the filter and a small 5x5x3 chunk of the image
(i.e. 5*5*3 = 75-dimensional dot product + bias)

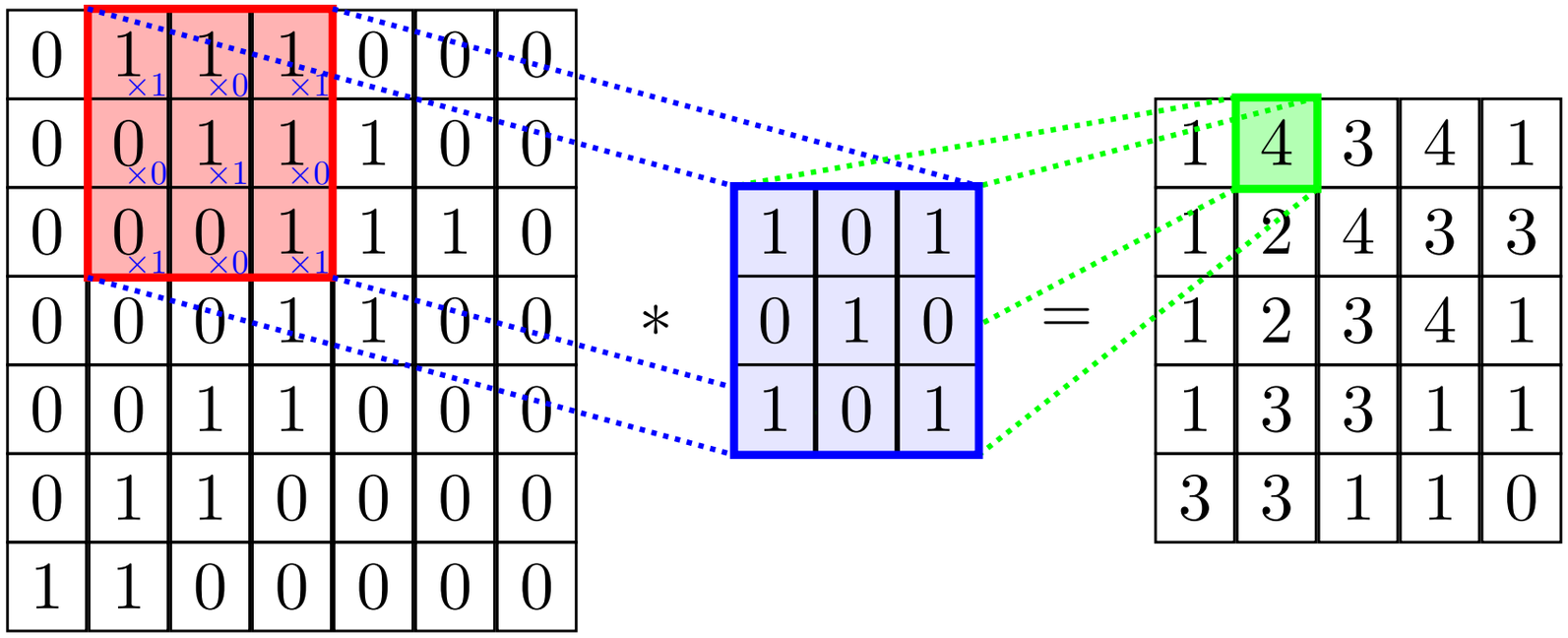
Convolution layers



$X_{n\times n}$ K Y

$$
\begin{aligned}
Y[1,1] = {} & X[1,1] * K[1,1] + X[1,2] * K[1,2] + X[1,3] * K[1,3] \\
& + X[2,1] * K[2,1] + X[2,2] * K[2,2] + X[2,3] * K[2,3] \\
& + X[3,1] * K[3,1] + X[3,2] * K[3,2] + X[3,3] * K[3,3]
\end{aligned}
$$

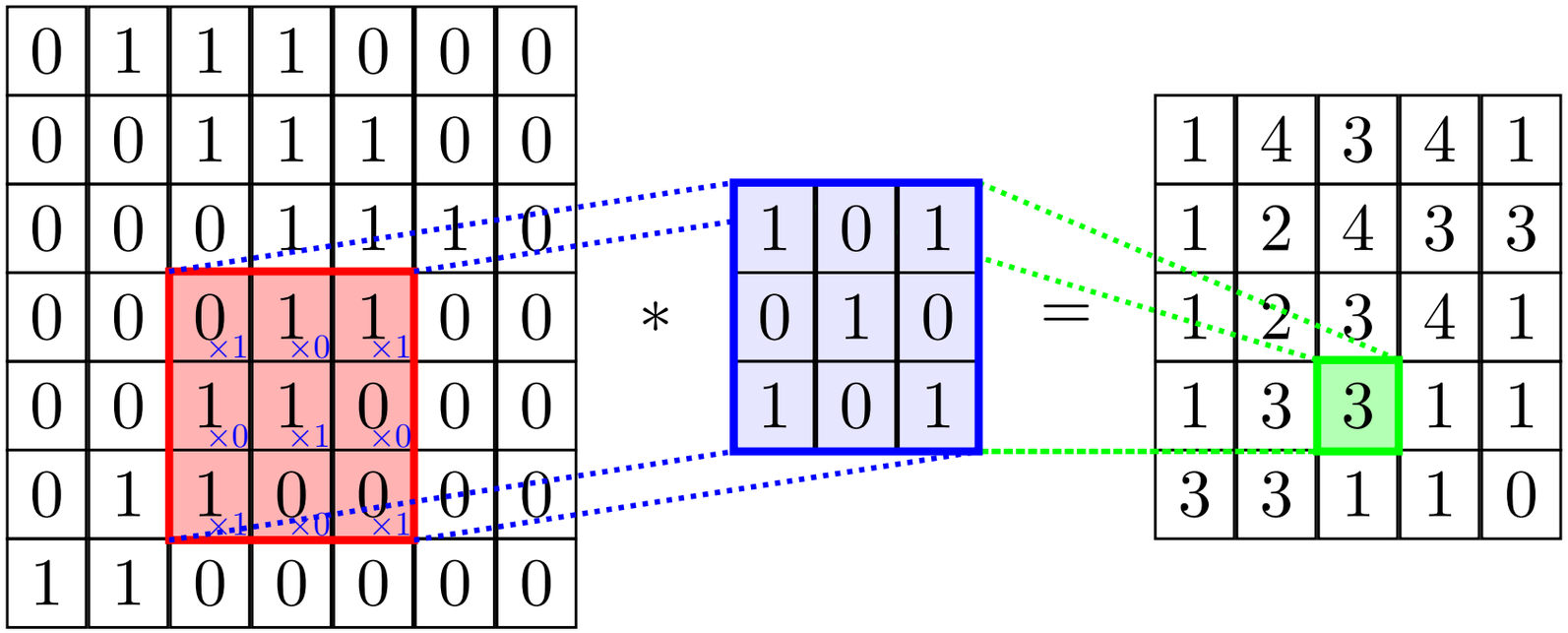
Convolution layers



$X_{n\times n}$ K Y

$$
\begin{aligned}
Y[1,2] = & X[1,2] * K[1,1] + X[1,3] * K[1,2] + X[1,4] * K[1,3] \\
& + X[2,2] * K[2,1] + X[2,3] * K[2,2] + X[2,4] * K[2,3] \\
& + X[3,2] * K[3,1] + X[3,3] * K[3,2] + X[3,4] * K[3,3]
\end{aligned}
$$

Convolution layers

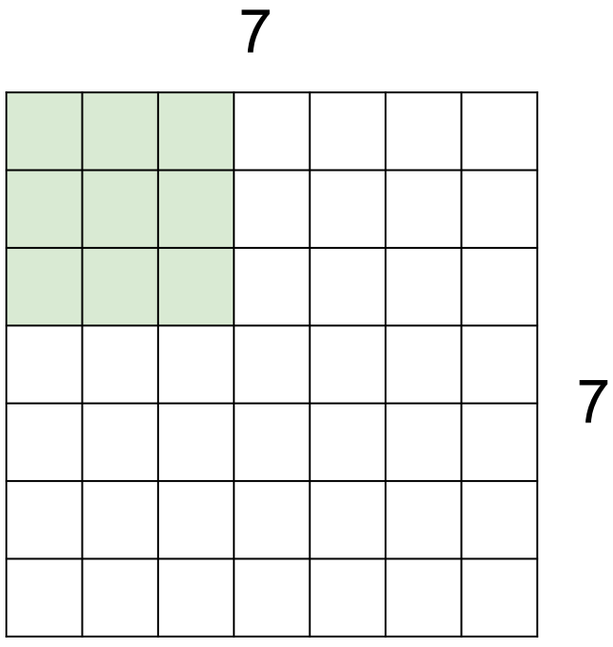


$X_{n\times n}$ K Y

$$Y[i,j] = \sum_{k_1=1}^{n}\sum_{k_2=1}^{n} X[k_1,k_2]K[i-k_1,j-k_2]$$

$$K[u,q] = 0 \qquad u > 3, q > 3$$

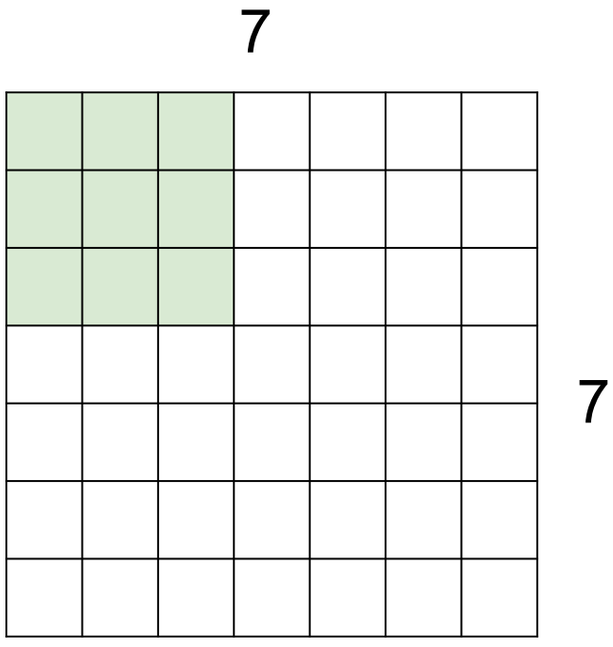
A closer look to spatial dimensions. Stride & Zero padding



7x7 input

Stride: Filter Step Size

A closer look to spatial dimensions. Stride & Zero padding

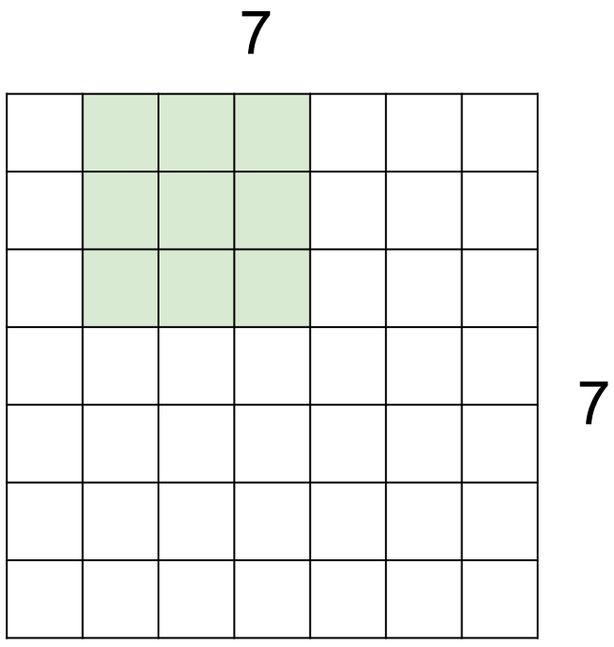


7x7 input

Stride: Filter Step Size

Stride = 1

A closer look to spatial dimensions. Stride & Zero padding

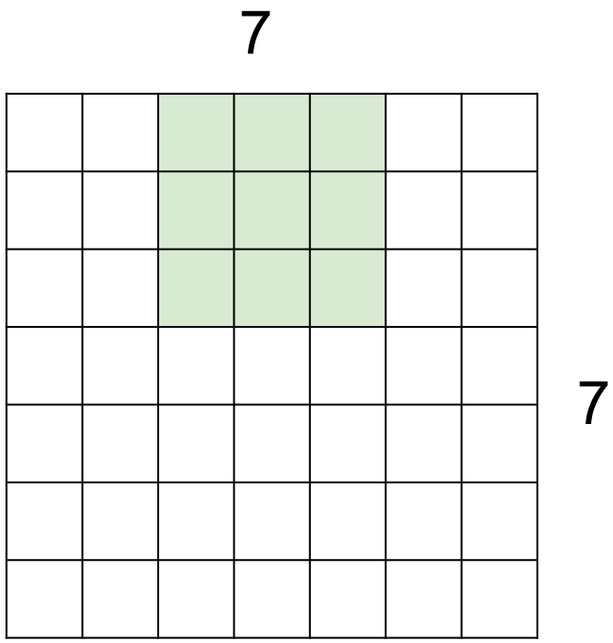


7x7 input

Stride: Filter Step Size

Stride = 1

A closer look to spatial dimensions. Stride & Zero padding

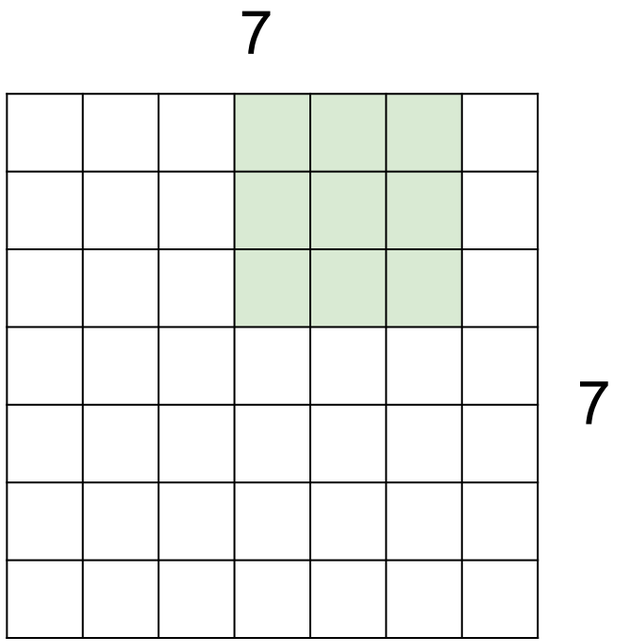


7x7 input

Stride: Filter Step Size

Stride = 1

A closer look to spatial dimensions. Stride & Zero padding

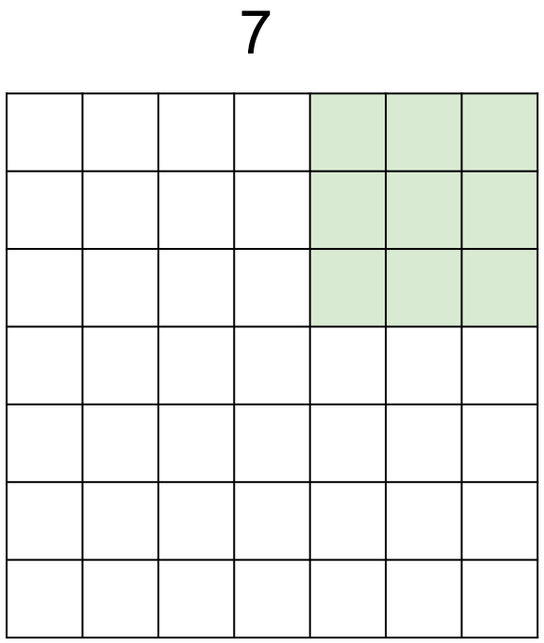


7x7 input

Stride: Filter Step Size

Stride = 1

A closer look to spatial dimensions. Stride & Zero padding

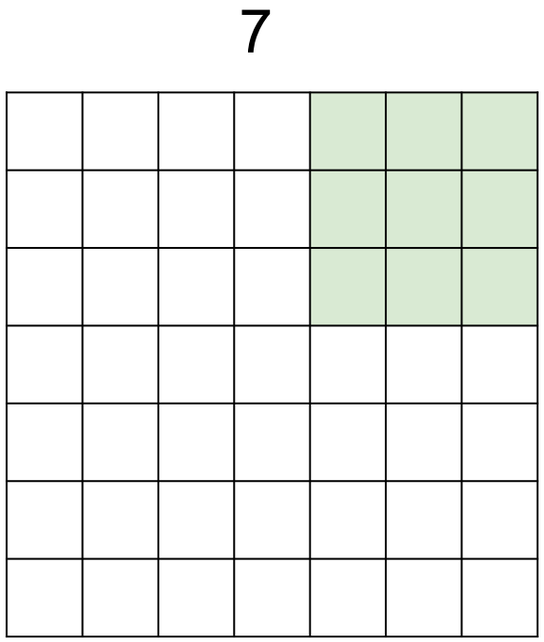


7x7 input

Stride: Filter Step Size

Stride = 1

A closer look to spatial dimensions. Stride & Zero padding



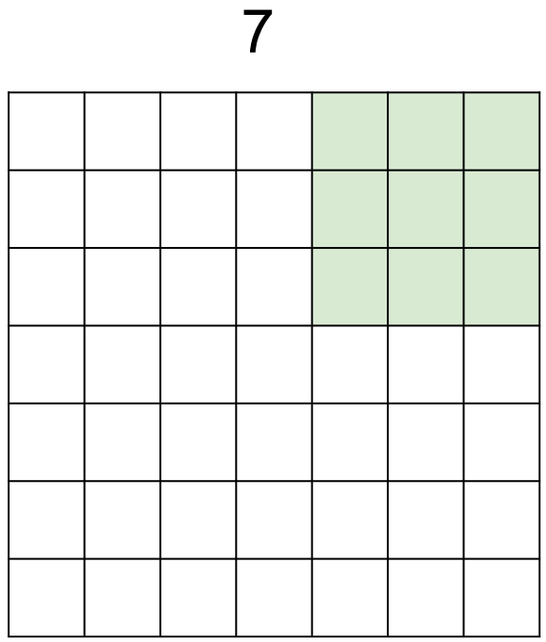
7x7 input

Stride: Filter Step Size

Stride = 1

5x5 Output

A closer look to spatial dimensions. Stride & Zero padding

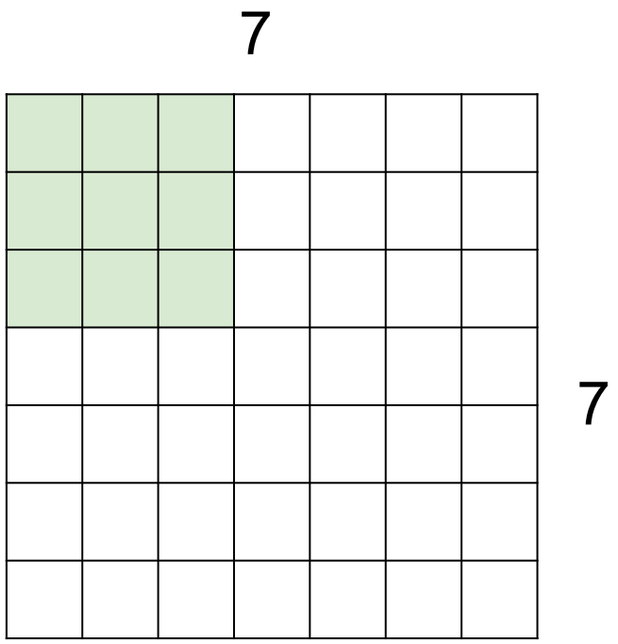


7x7 input

Stride: Filter Step Size

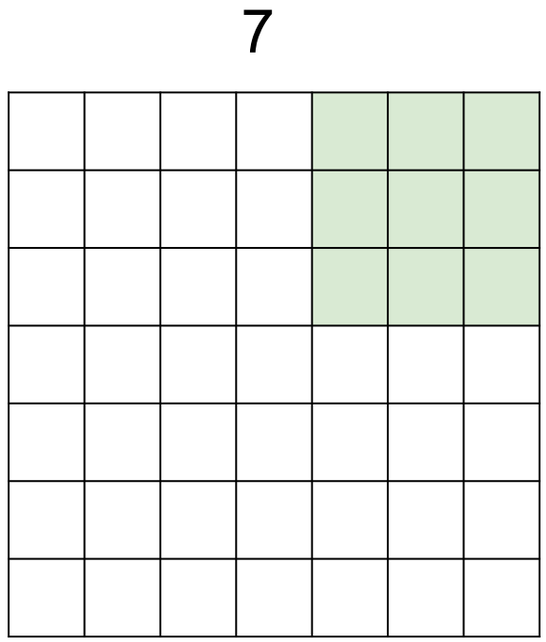
Stride = 1

5x5 Output



Stride = 2

A closer look to spatial dimensions. Stride & Zero padding

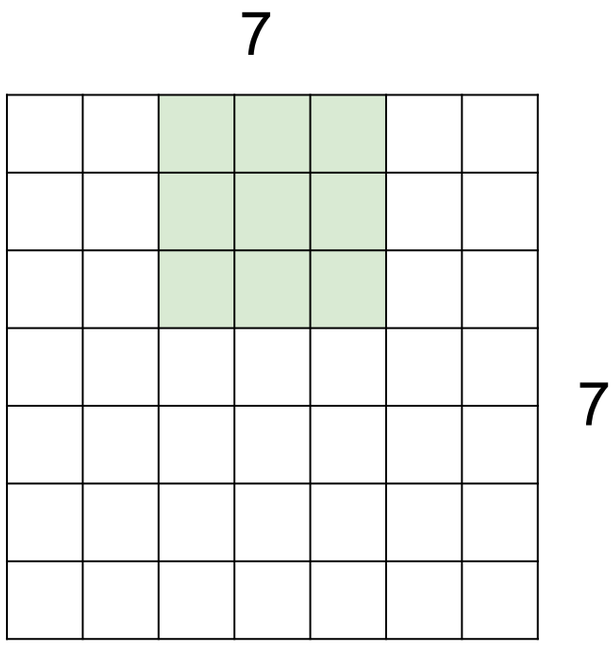


7x7 input

Stride: Filter Step Size

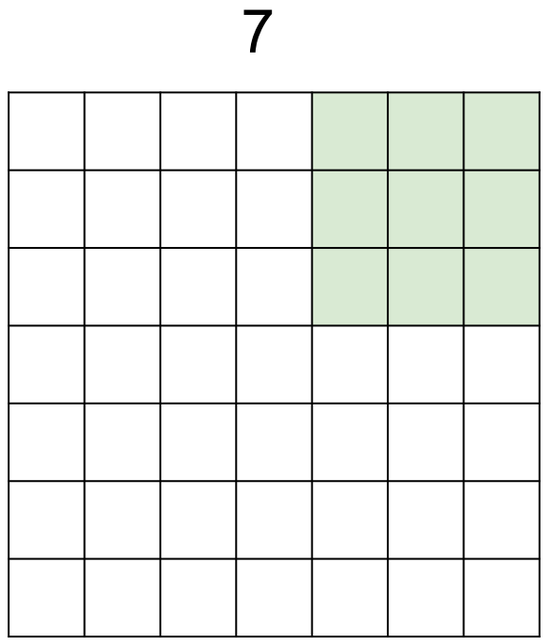
Stride = 1

5x5 Output



Stride = 2

A closer look to spatial dimensions. Stride & Zero padding

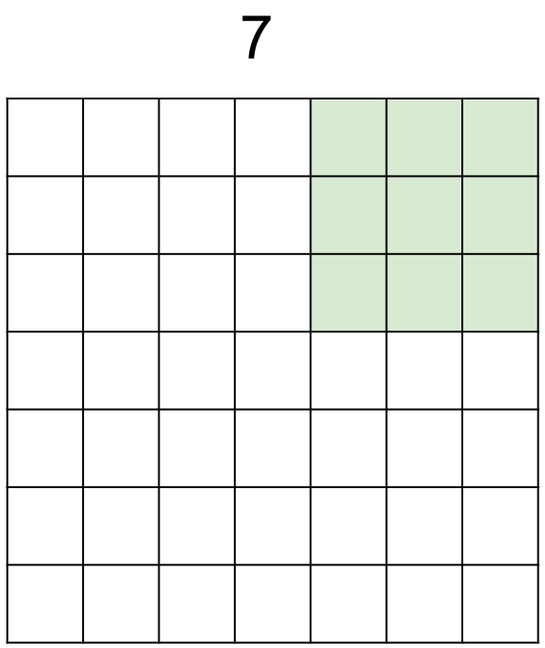


7x7 input

Stride: Filter Step Size

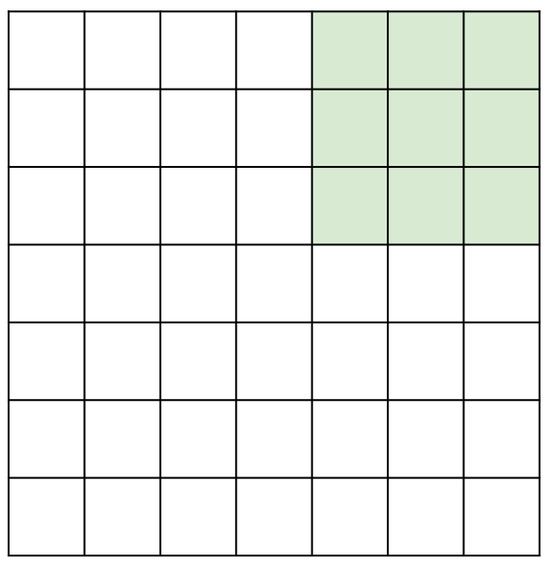
Stride = 1

5x5 Output



Stride = 2

A closer look to spatial dimensions. Stride & Zero padding

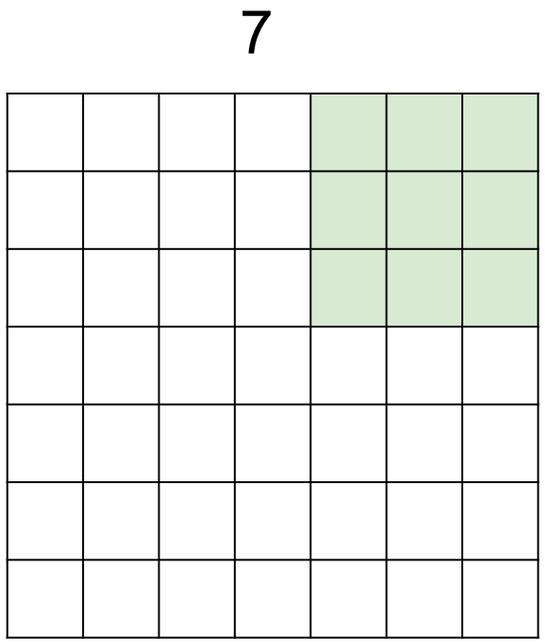


7x7 input

Stride: Filter Step Size

Stride = 1

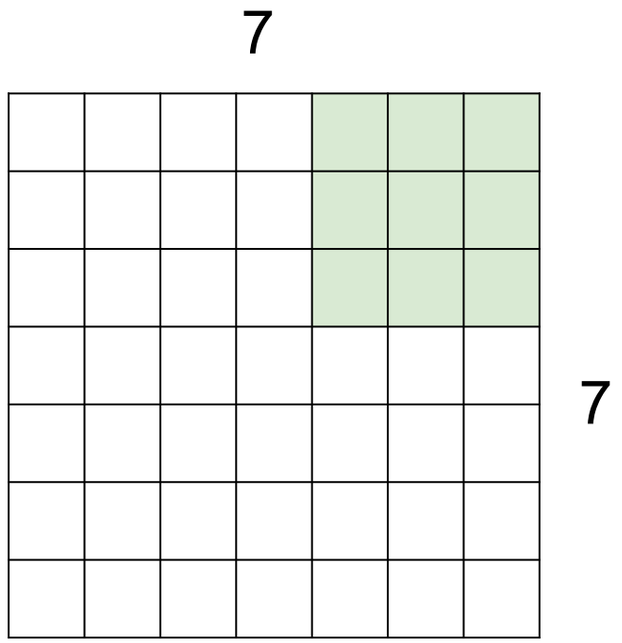
5x5 Output



Stride = 2

3x3 Output

A closer look to spatial dimensions. Stride & Zero padding

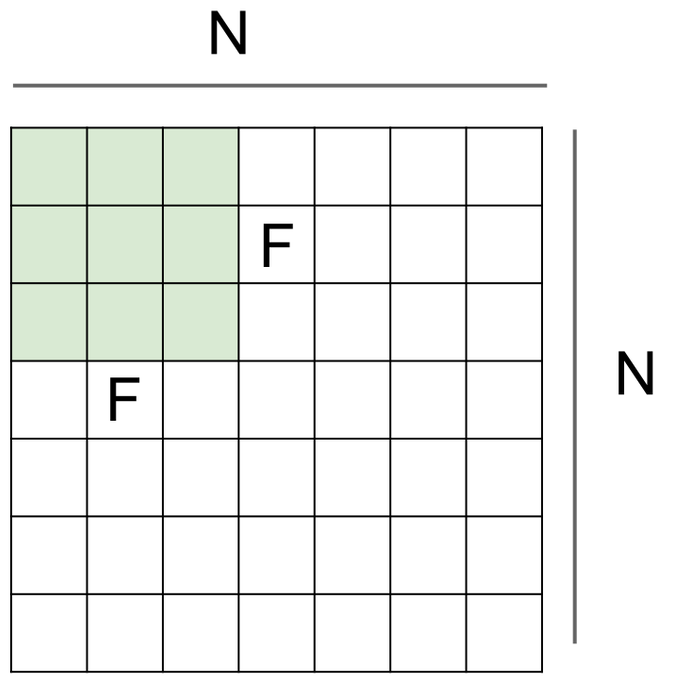


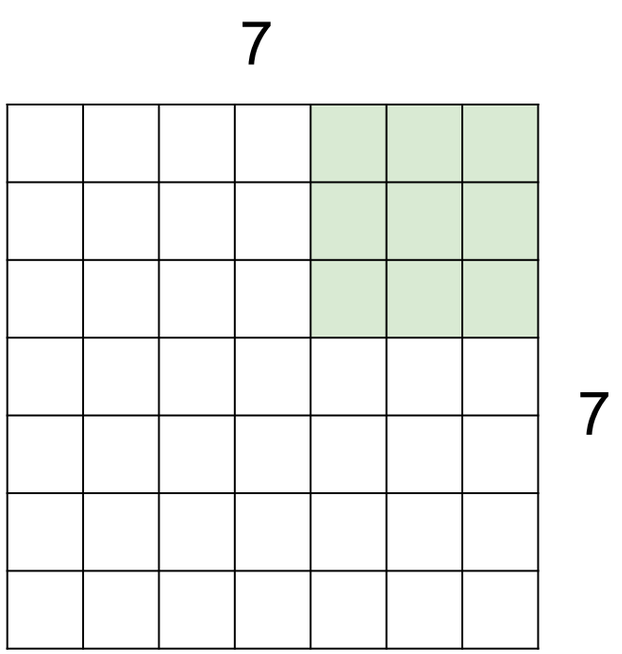
7x7 input

Stride: Filter Step Size

Stride = 1

5x5 Output





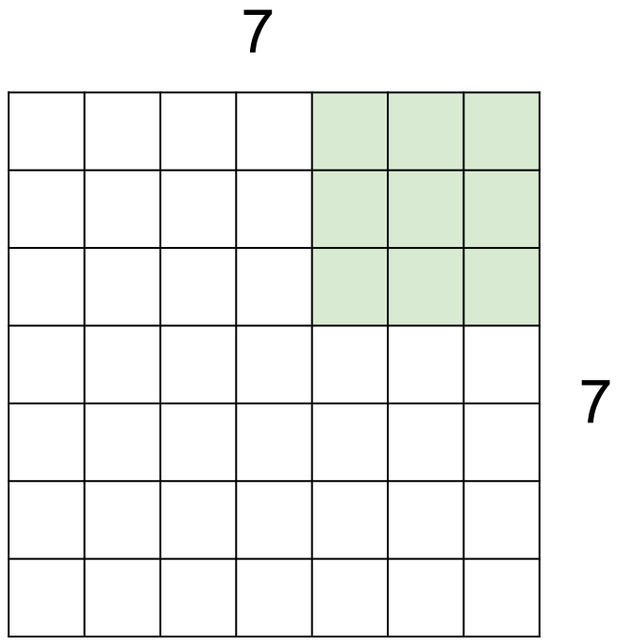
Stride = 2

3x3 Output

Output size:

$$(N - F)/\text{stride} + 1$$

A closer look to spatial dimensions. Stride & Zero padding

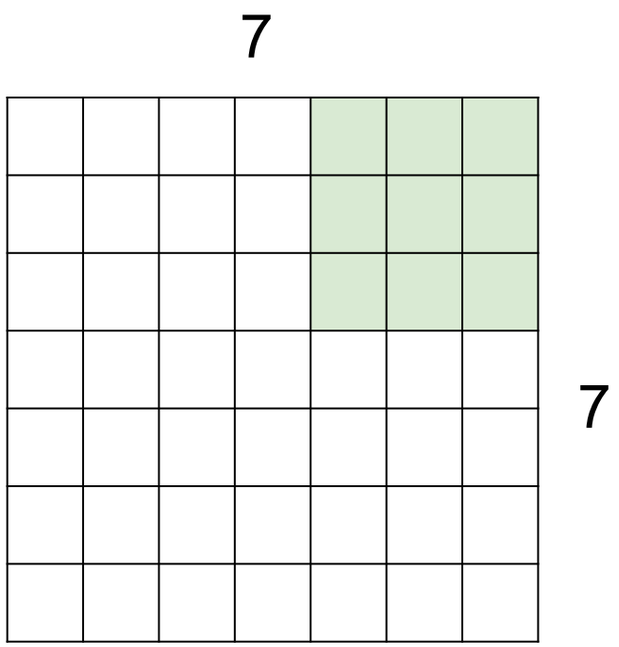


7x7 input

Stride: Filter Step Size

Stride = 1

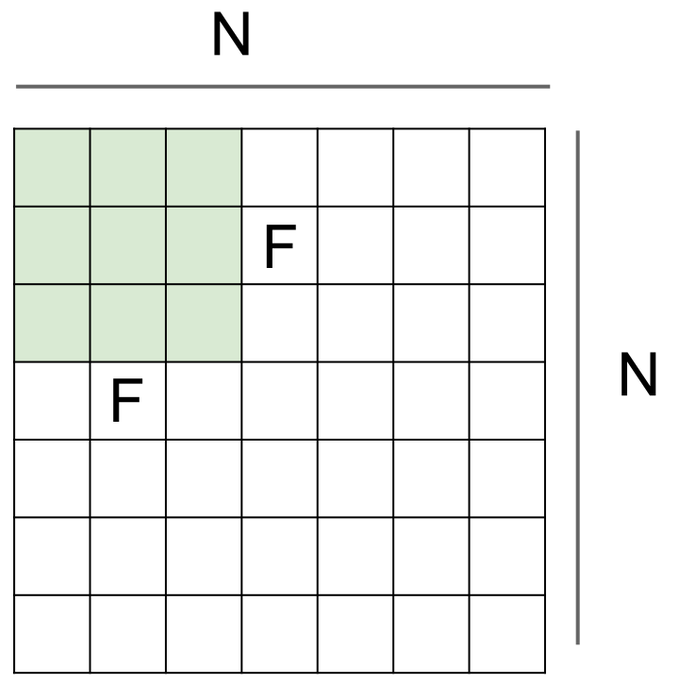
5x5 Output



Stride = 2

3x3 Output

In practice: zero-pad the image when needed



Output size:

$$(N - F)/\text{stride} + 1$$

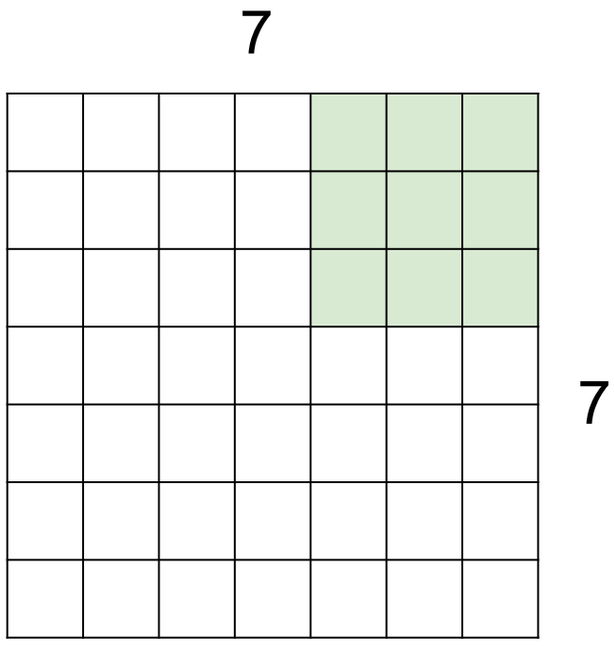
A closer look to spatial dimensions. Stride & Zero padding

7x7 input

Stride: Filter Step Size

Stride = 1

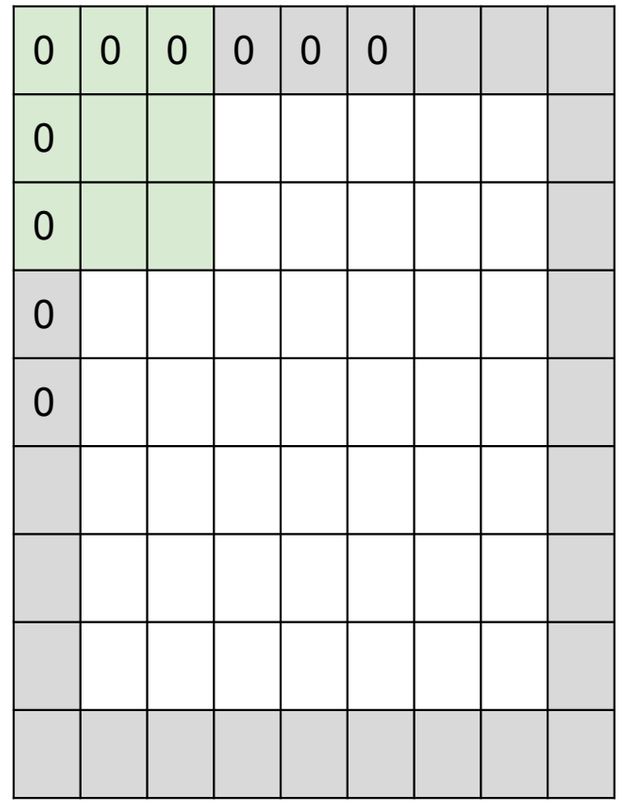
5x5 Output



Stride = 2

3x3 Output

In practice: zero-pad the image when needed

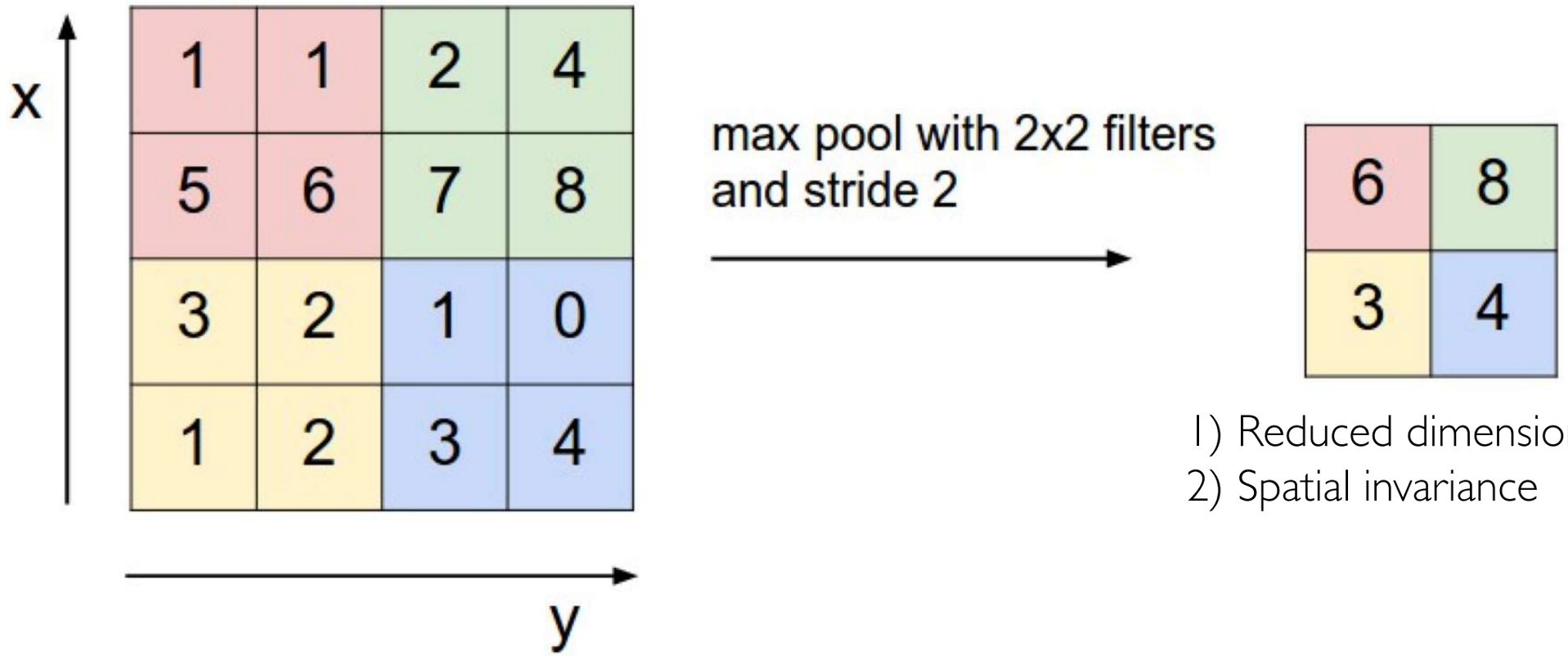


Output size:

$$(N - F)/\text{stride} + 1$$

Pooling Layers

Alternatively to stride >1, we can use *pooling layers*



1) Reduced dimensionality
2) Spatial invariance

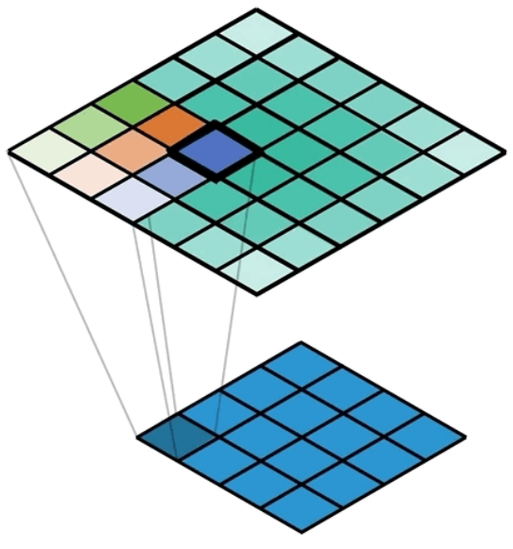
Biggest reason use of pooling is historical.
Common pretrained models use pooling, so finetuned and derived models use pooling too.

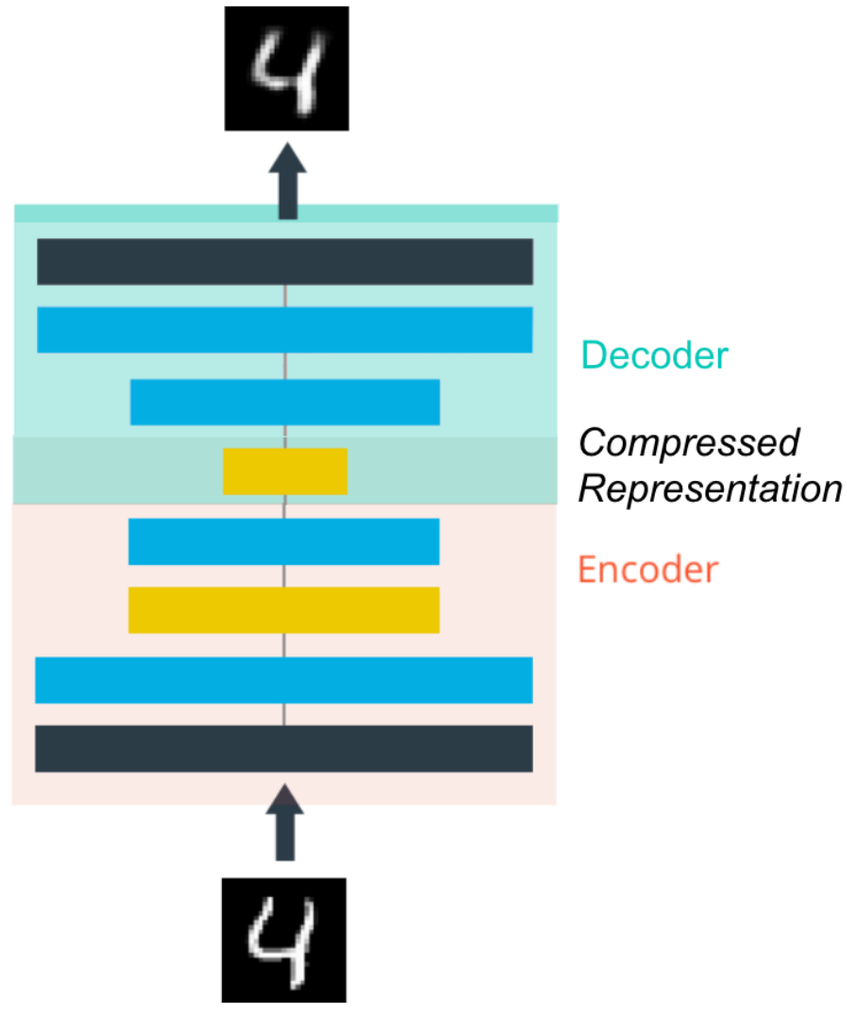
Autoencoders for images. Transpose convolution layers

- The transposed convolution operation forms the same connectivity as the normal convolution but in the **backward direction**

- We can use it to **conduct up-sampling**

- The **weights in the transposed convolution are learnable**. So we do not need a predefined interpolation method.



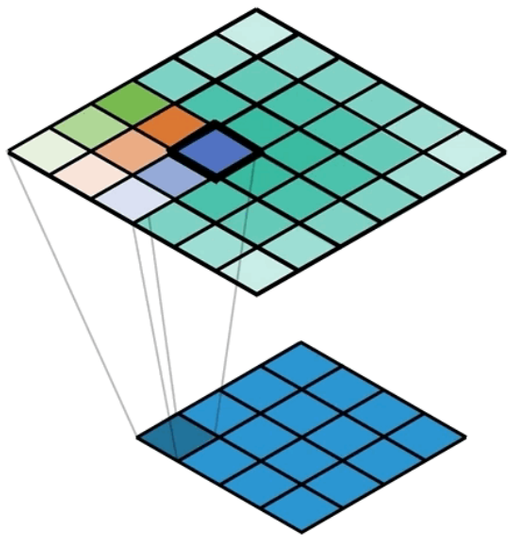


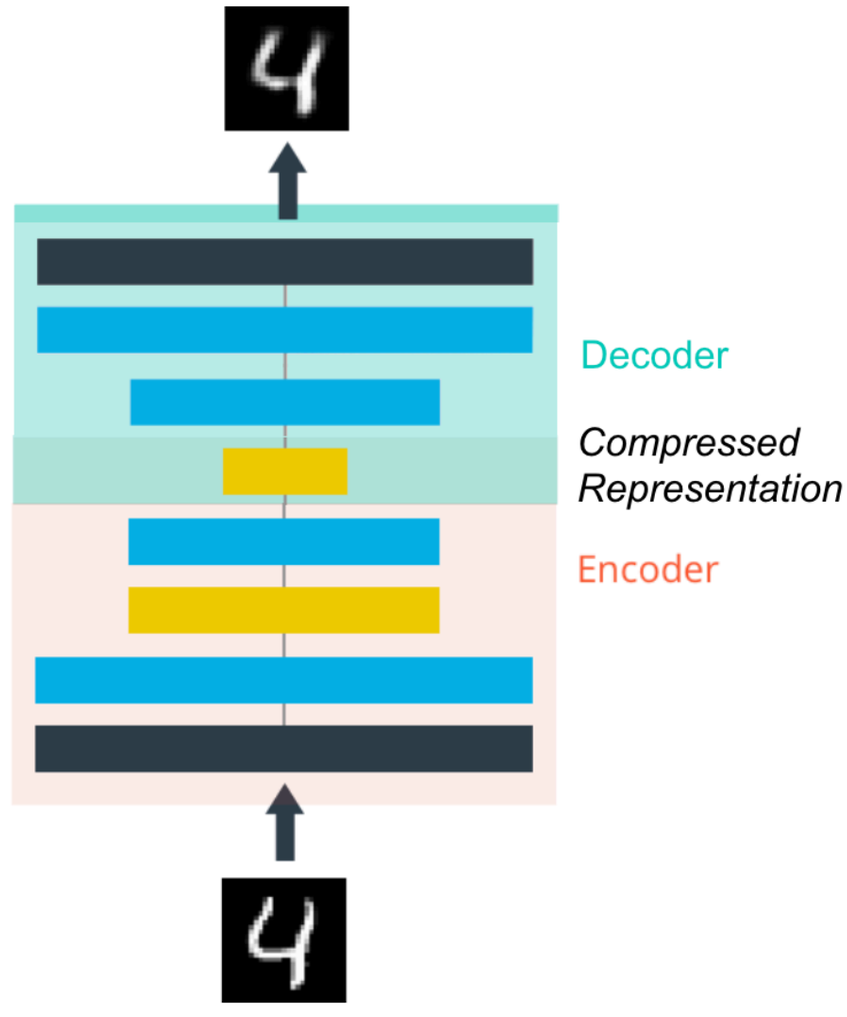
Autoencoders for images. Transpose convolution layers

- The transposed convolution operation forms the same connectivity as the normal convolution but in the **backward direction**

- We can use it to **conduct up-sampling**

- The **weights in the transposed convolution are learnable**. So we do not need a predefined interpolation method.



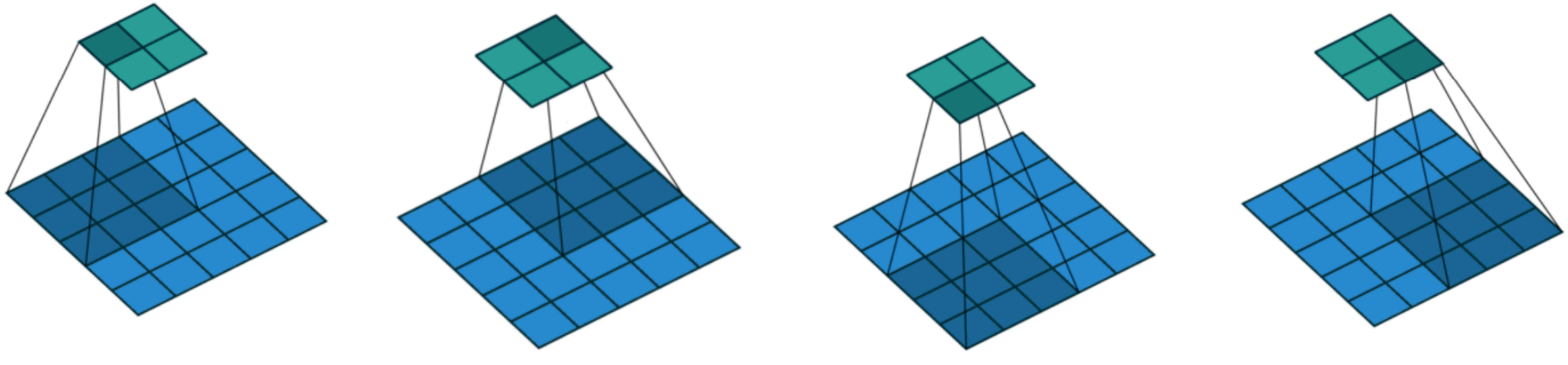


Comparison between convolution and transposed convolution

- Regular convolution:

output

input

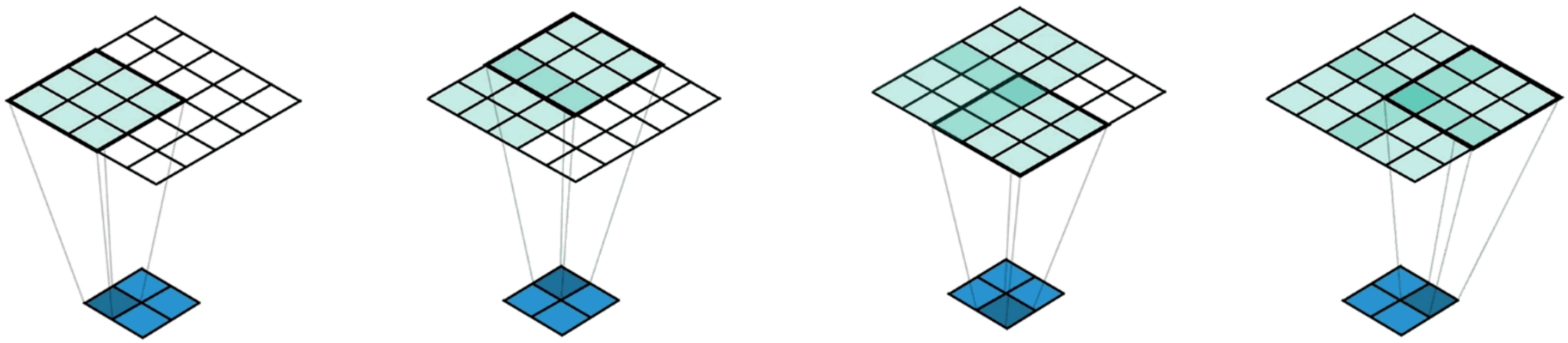


Dumoulin, Vincent, and Francesco Visin. "A guide to convolution arithmetic for deep learning." arXiv preprint arXiv:1603.07285 (2016).

- Transposed convolution (stride 2)

output

input



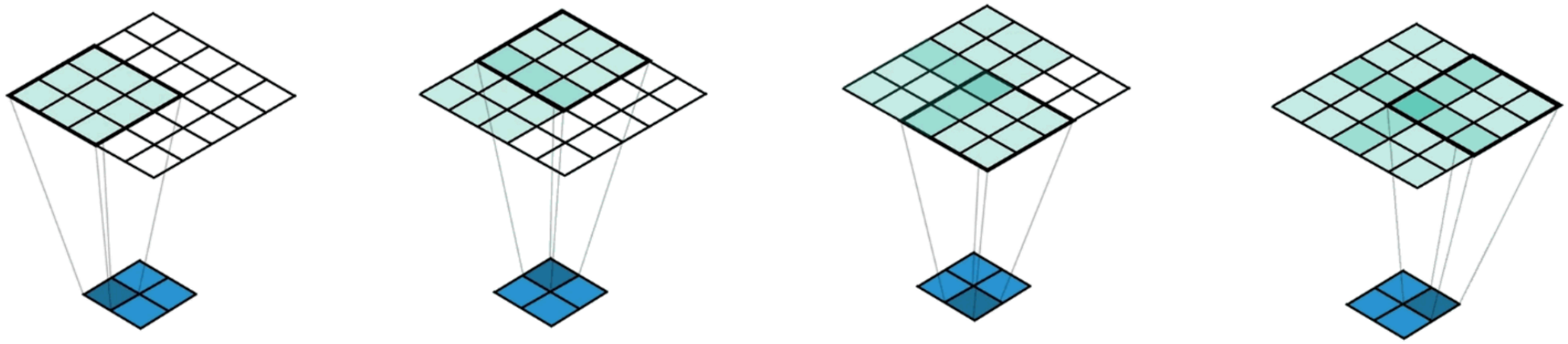
A Conv2DTranspose with 3x3 kernel and stride of 2x2 applied to a 2x2 input to give a 5x5 output.
(https://medium.com/apache-mxnet/transposed-convolutions-explained-with-ms-excel-52d13030c7e8)

Comparison between convolution and transposed convolution

- Transposed Convolution (3x3 kernel, stride=2):

output

input

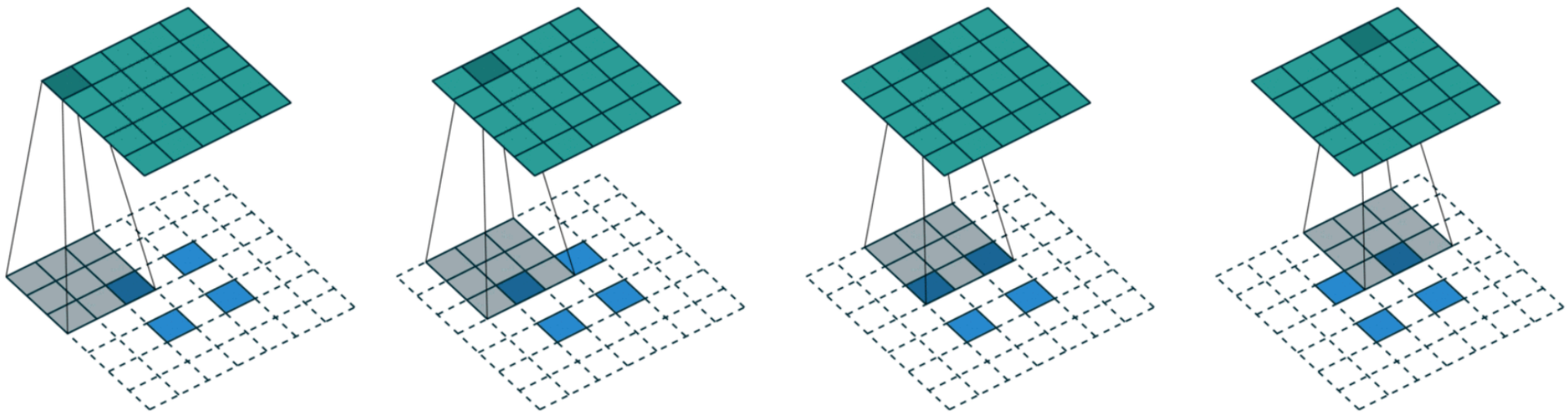


A Conv2DTranspose with 3x3 kernel and stride of 2x2 applied to a 2x2 input to give a 5x5 output.
(https://medium.com/apache-mxnet/transposed-convolutions-explained-with-ms-excel-52d13030c7e8)

- Transposed convolution (emulated with direct convolution)

output

input



Dumoulin, Vincent, and Francesco Visin. "A guide to convolution arithmetic for deep learning." arXiv preprint arXiv:1603.07285 (2016).

Comparison between convolution and transposed convolution

- Regular convolution (stride 1):

output

input

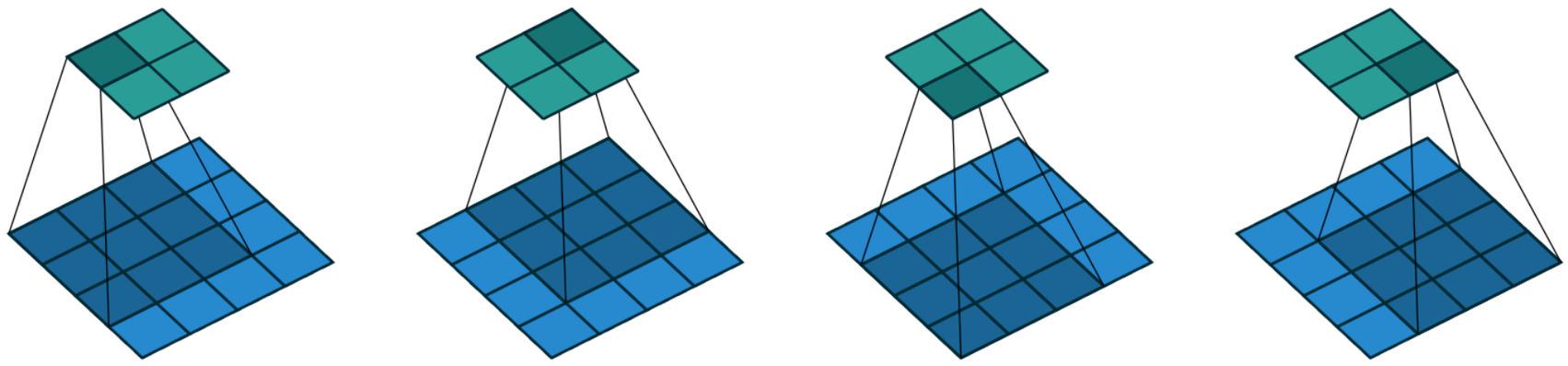
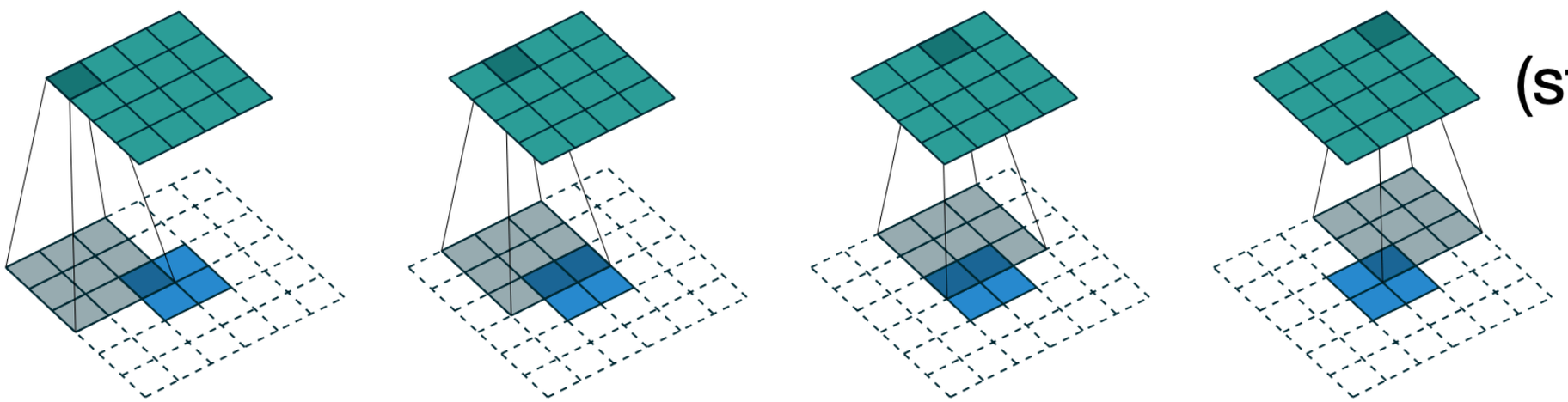


Figure 2.1: (No padding, unit strides) Convolving a 3×3 kernel over a 4×4 input using unit strides (i.e., $i = 4$, $k = 3$, $s = 1$ and $p = 0$).

- Transposed convolution (emulated with the direct convolution)

output

input

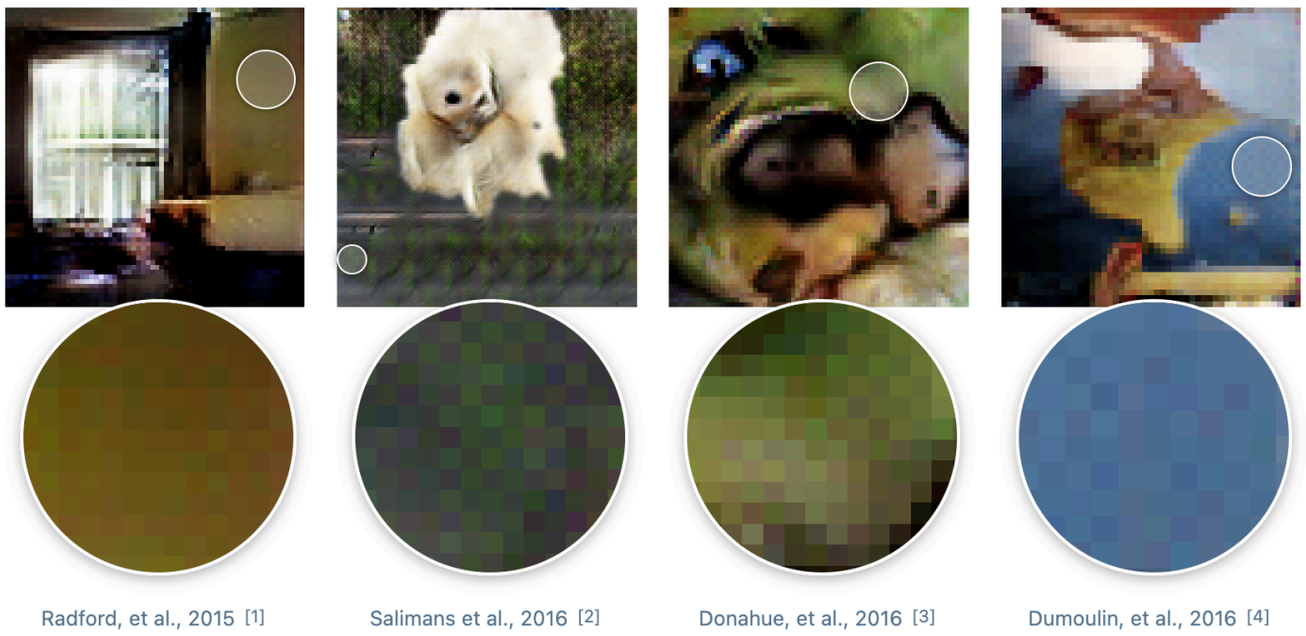


Dumoulin, Vincent, and Francesco Visin. "A guide to convolution arithmetic for deep learning." arXiv preprint arXiv:1603.07285 (2016).

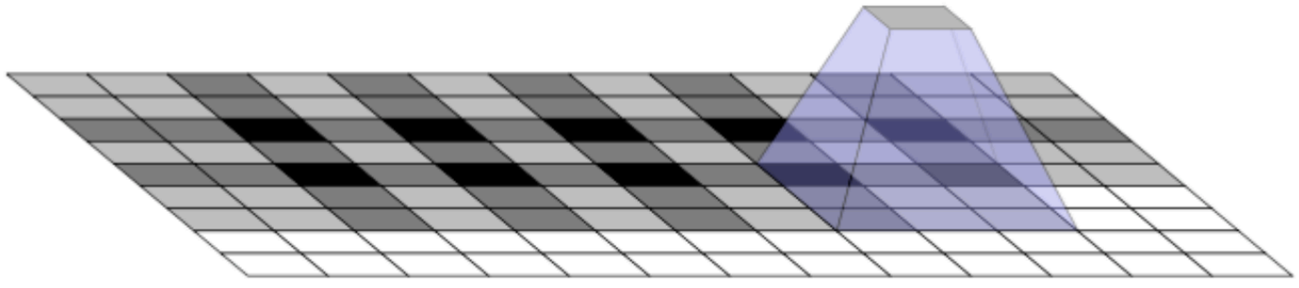
Deconvolution and checkboard artifacts

When we look very closely at images generated by neural networks, we often see a strange checkerboard pattern of artifacts. It's more obvious in some cases than others, but a large fraction of recent models exhibit this behavior.

Link

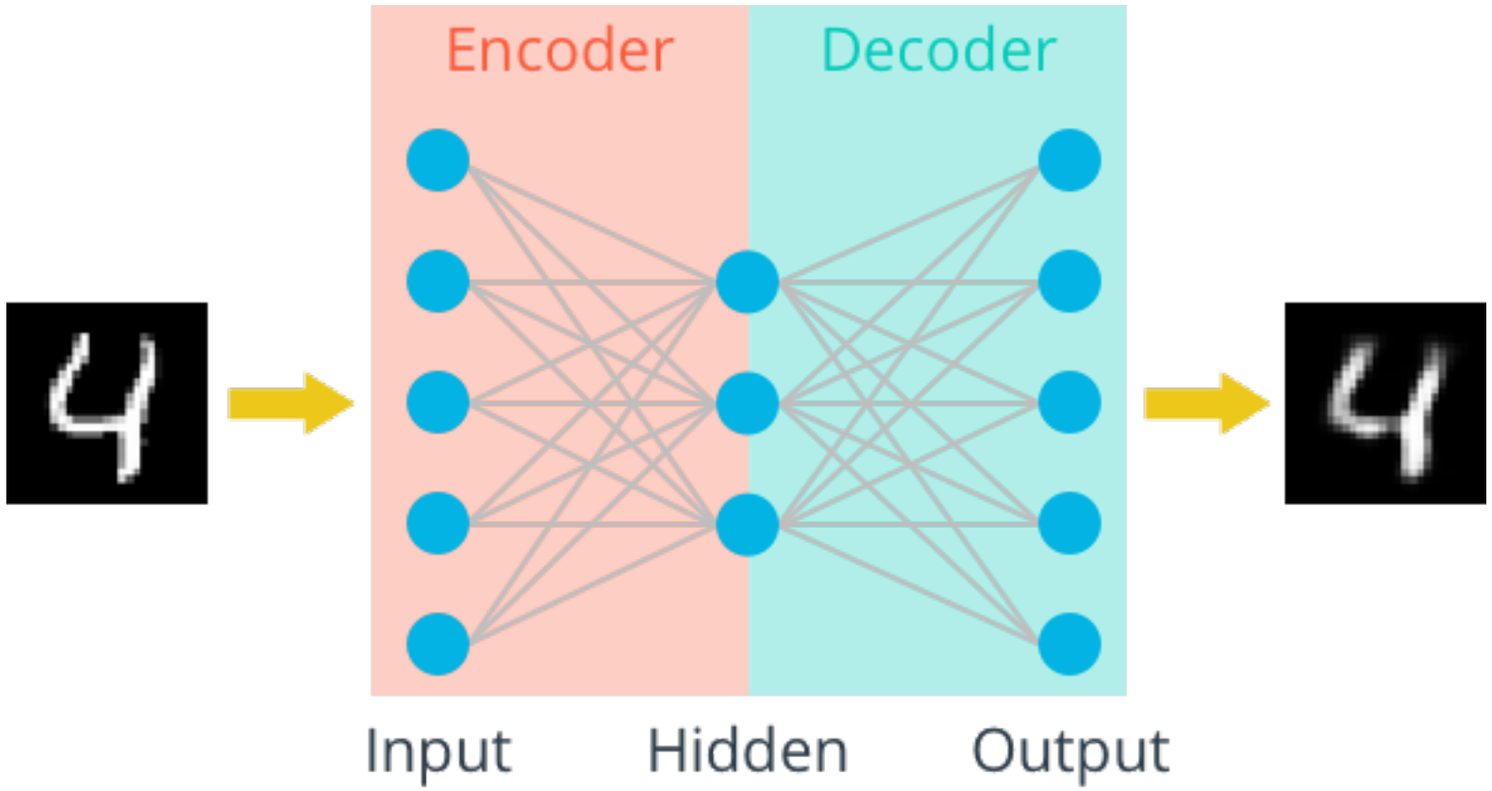


Radford, et al., 2015 [1] Salimans et al., 2016 [2] Donahue, et al., 2016 [3] Dumoulin, et al., 2016 [4]



- **Recommendation**: Replace transposed convolution by
  upsampling interpolation + regular convolution

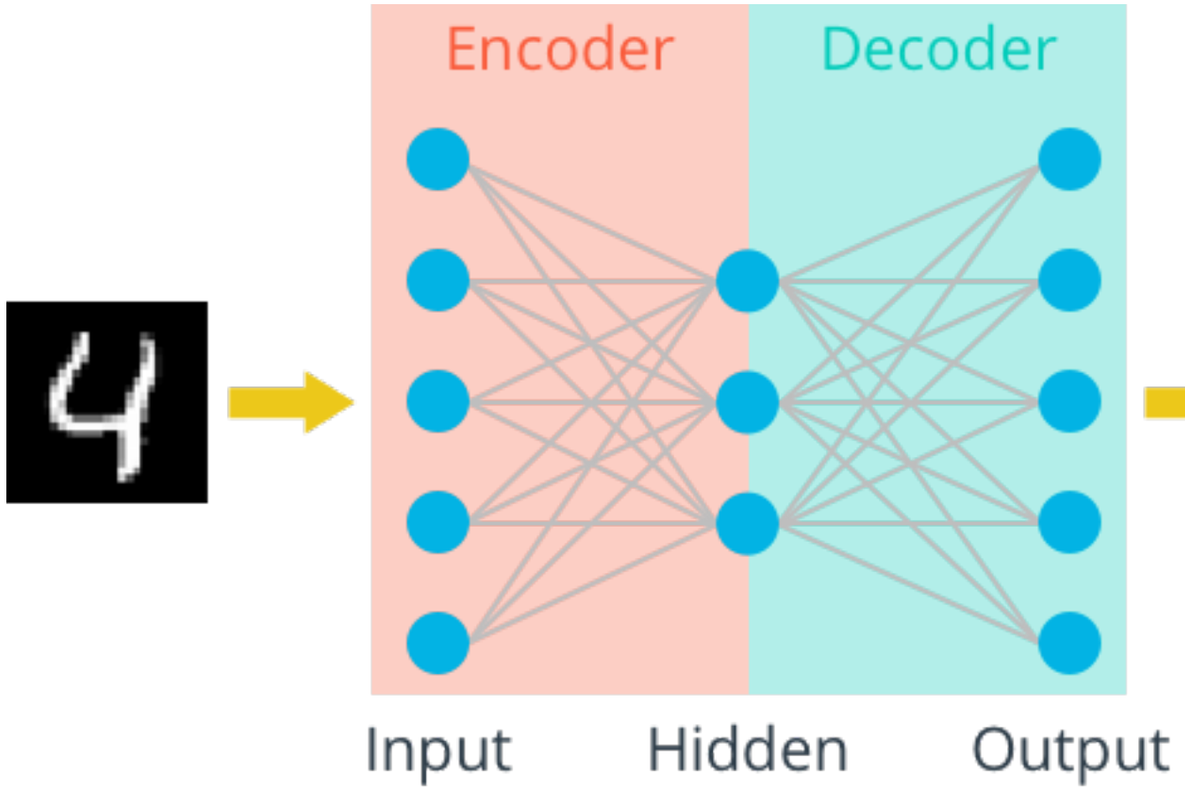
Sparse Deep Autoencoders



$$L(\boldsymbol{x}, g(f(\boldsymbol{x}))) + \Omega(\boldsymbol{h})$$

- Sparse autoencoders are typically used to learn features for another task

Sparse Deep Autoencoders



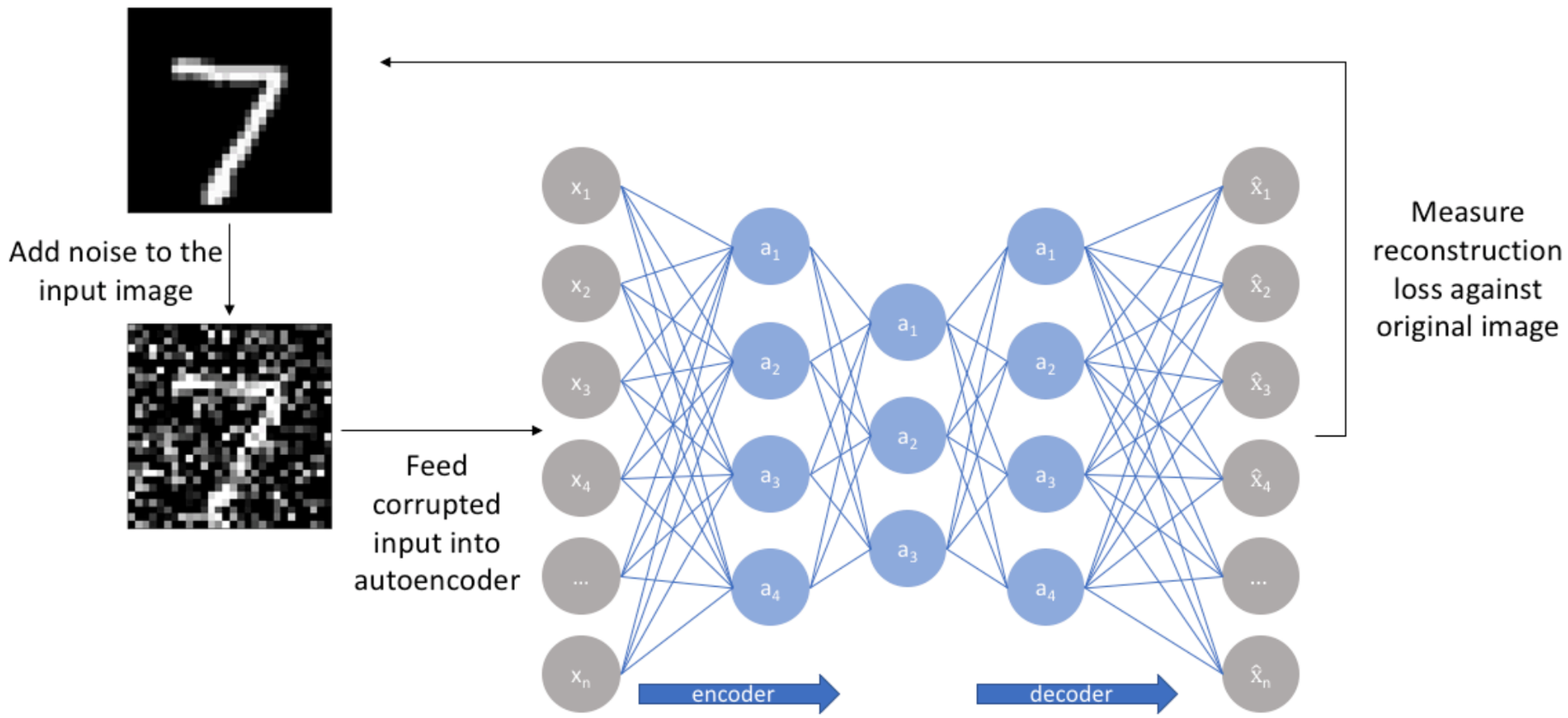
$$L(\boldsymbol{x}, g(f(\boldsymbol{x}))) + \Omega(\boldsymbol{h})$$

Sparsity penalizer

$$\Omega(\boldsymbol{h}) = \lambda \sum_i |h_i|$$

- Sparse autoencoders are typically used to learn features for another task

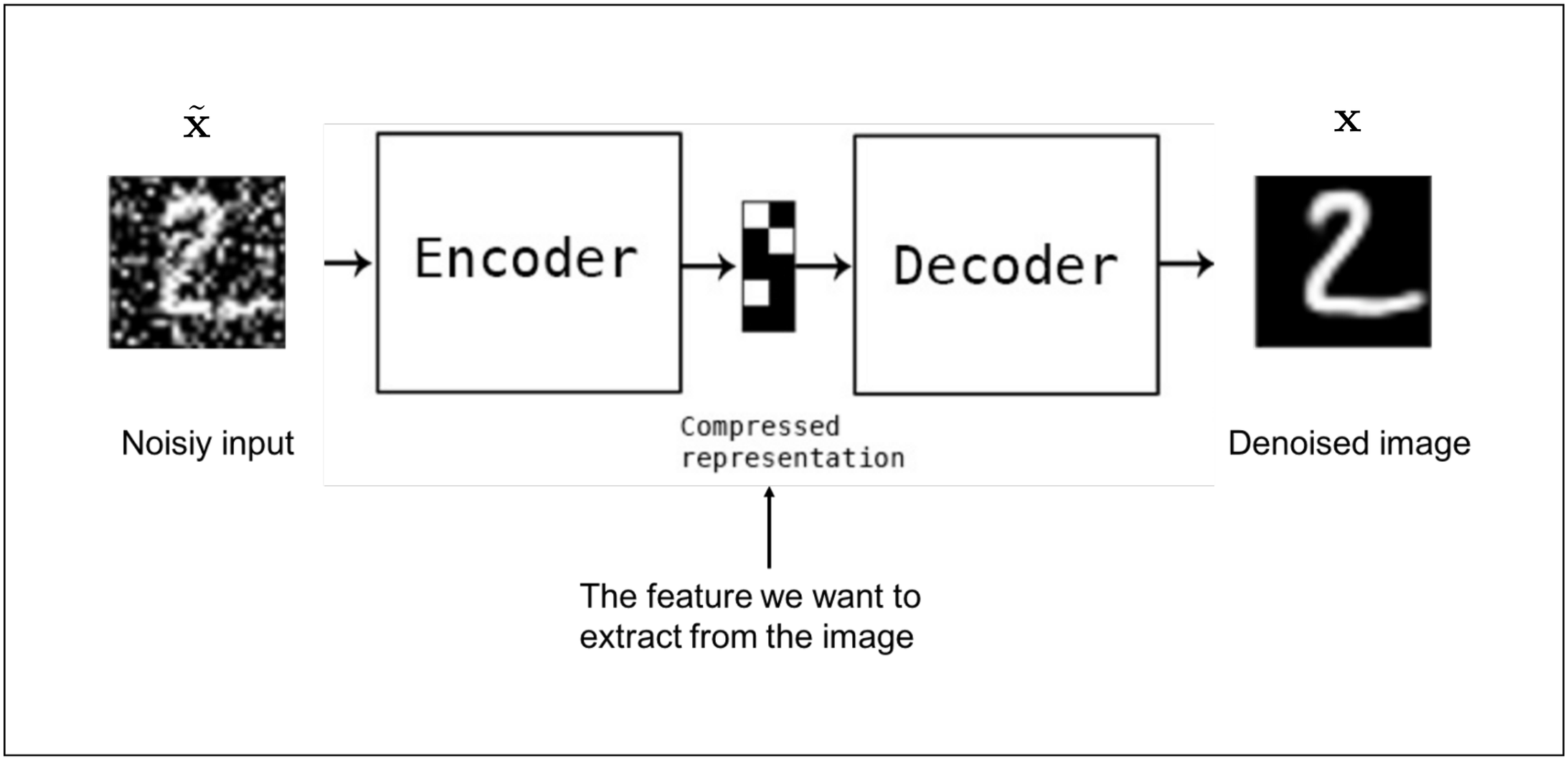
Denoising Deep Autoencoders



Source: this excellent blog

- Reconstruct a **corrupted version** of an image
- More robust solutions. It is some sort of regularization
- They are widely used for image denoising and missing data completion

Denoising Deep Autoencoders



$$\mathcal{L}(\eta, \theta) = \frac{1}{N} \sum_{n=1}^{N} \mathcal{L}_n \left(\mathbf{x}_n - D_\theta \left(E_\eta(\tilde{\mathbf{x}}_n) \right) \right)$$

Applications

- Dimensionality reduction
- Visualization

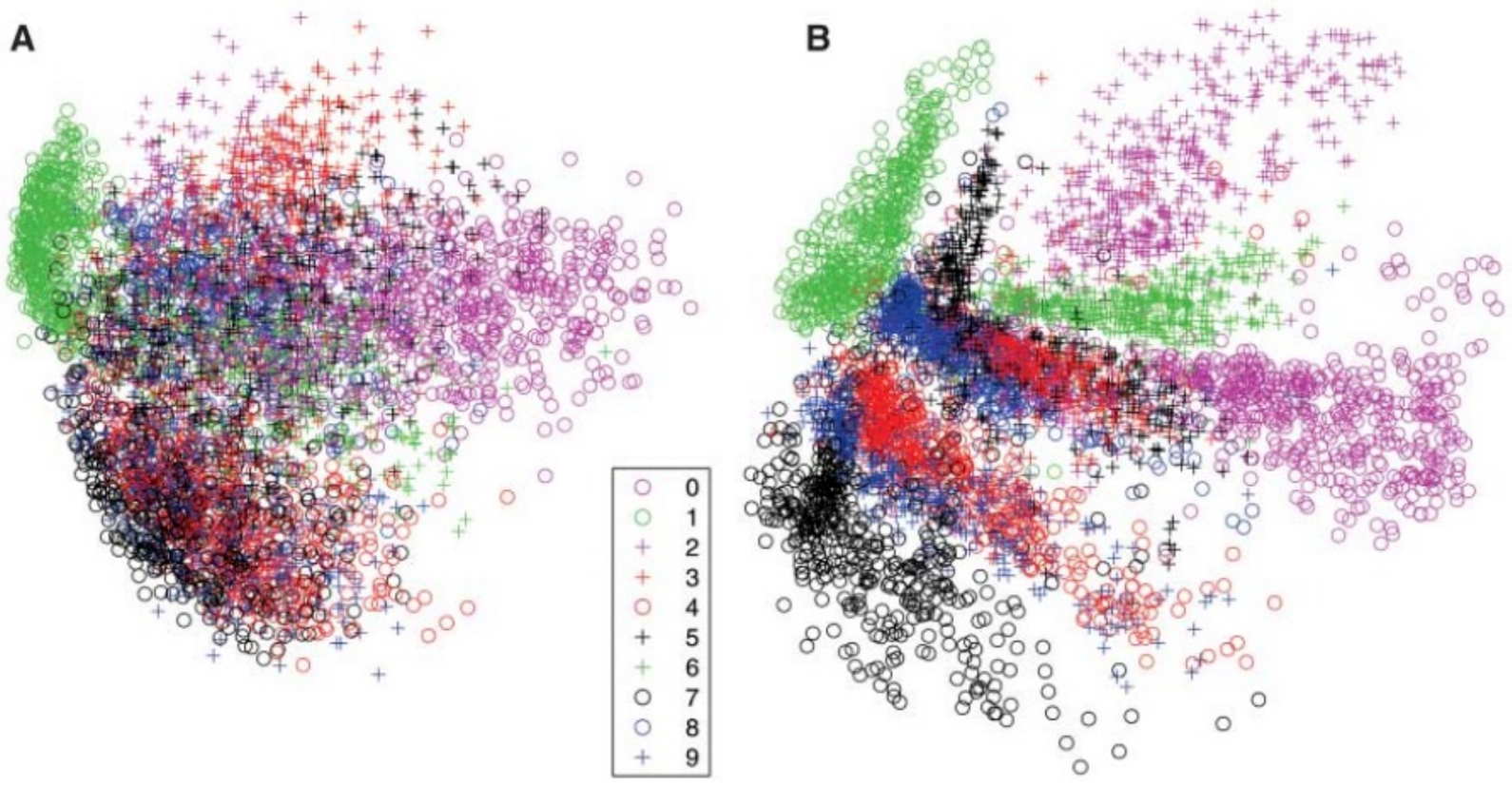
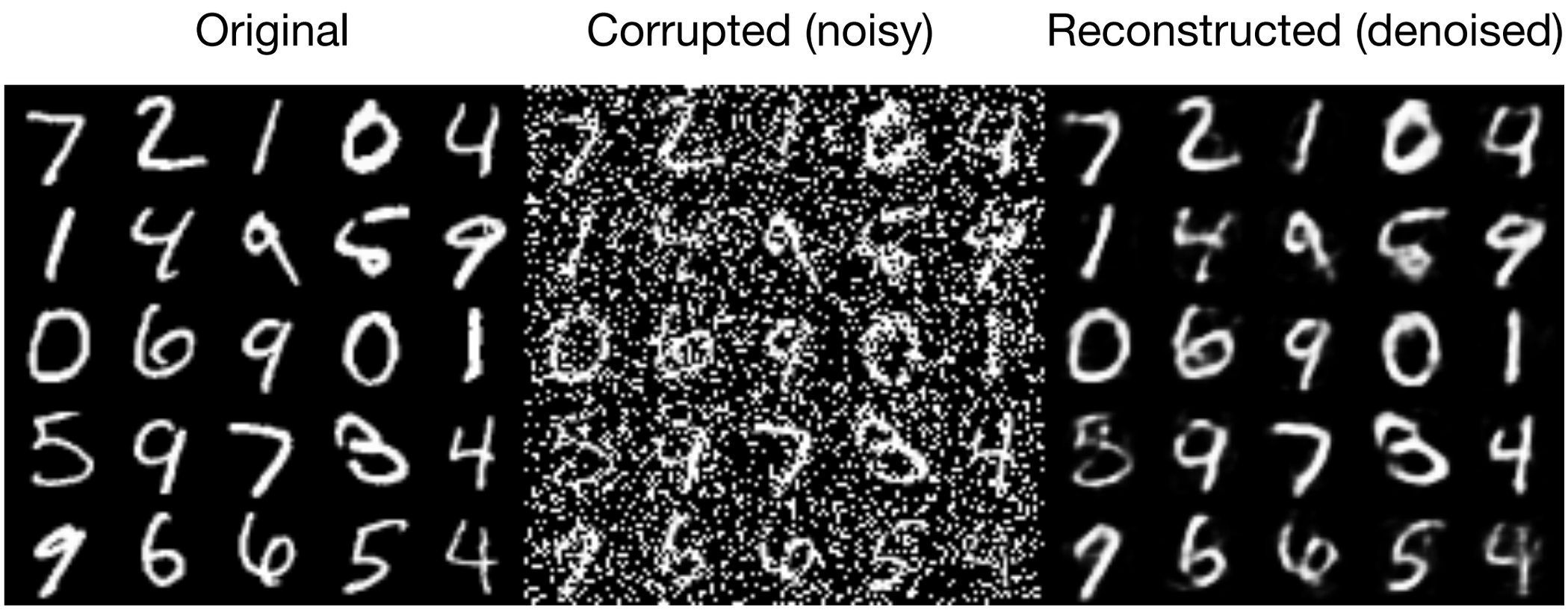


Fig. 3. (**A**) The two-dimensional codes for 500 digits of each class produced by taking the first two principal components of all 60,000 training images. (**B**) The two-dimensional codes found by a 784-1000-500-250-2 autoencoder. For an alternative visualization, see (*8*).

"Reducing the Dimensionality of Data with Neural Networks", G. E. Hinton and R. R. Salakhutdinov, Science, vol. 313, 2006.

Applications

- Denoising
- Missing data imputation



Applications

- Anomaly/outlier detection

95% of Driving Data:
(1) sunny, (2) highway, (3) straight road

Edge Cases

Harsh Weather

Pedestrians

© MIT 6.S191: Introduction to Deep Learning
introtodeeplearning.com

Image segmentation using CNNs

Image Segmentation Using Deep Learning: A Survey

Shervin Minaee, Yuri Boykov, Fatih Porikli, Antonio Plaza, Nasser Kehtarnavaz, and Demetri Terzopoulos

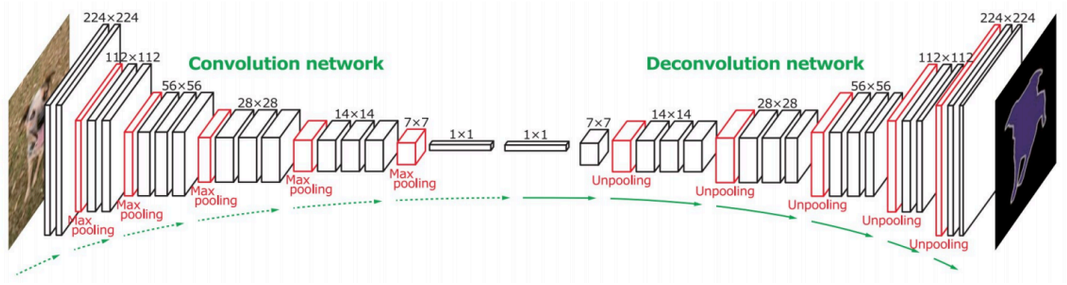


Fig. 11. Deconvolutional semantic segmentation. Following a convolution network based on the VGG 16-layer net, is a multi-layer deconvolution network to generate the accurate segmentation map. From [42].

- Most powerful methods are based on **encoder-decoder networks**

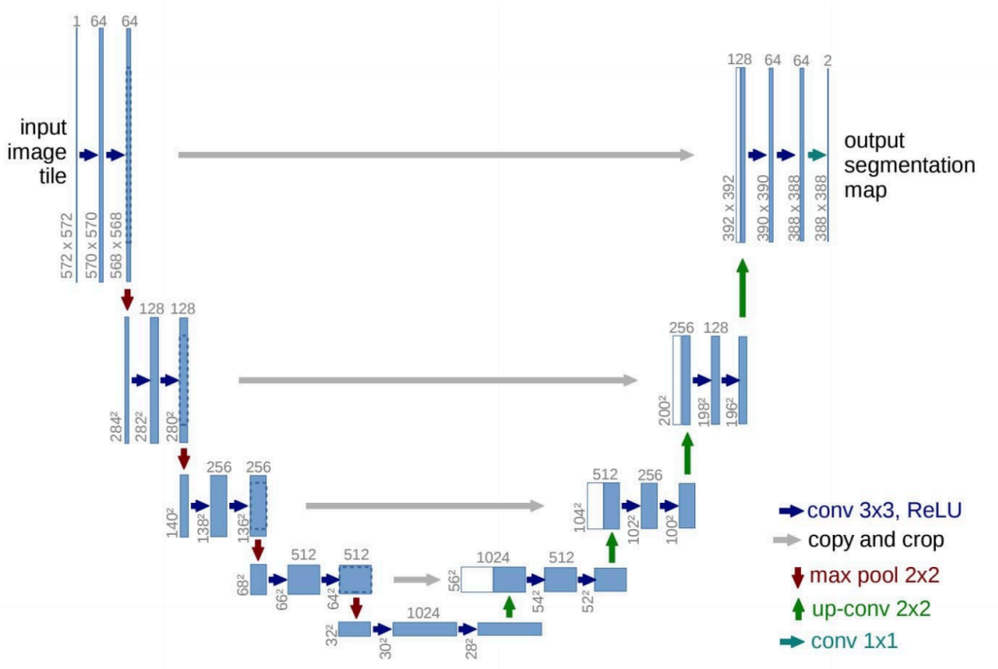


Fig. 14. The U-net model. The blue boxes denote feature map blocks with their indicated shapes. From [49].

Image segmentation using CNNs

- Image segmentation using CNNs

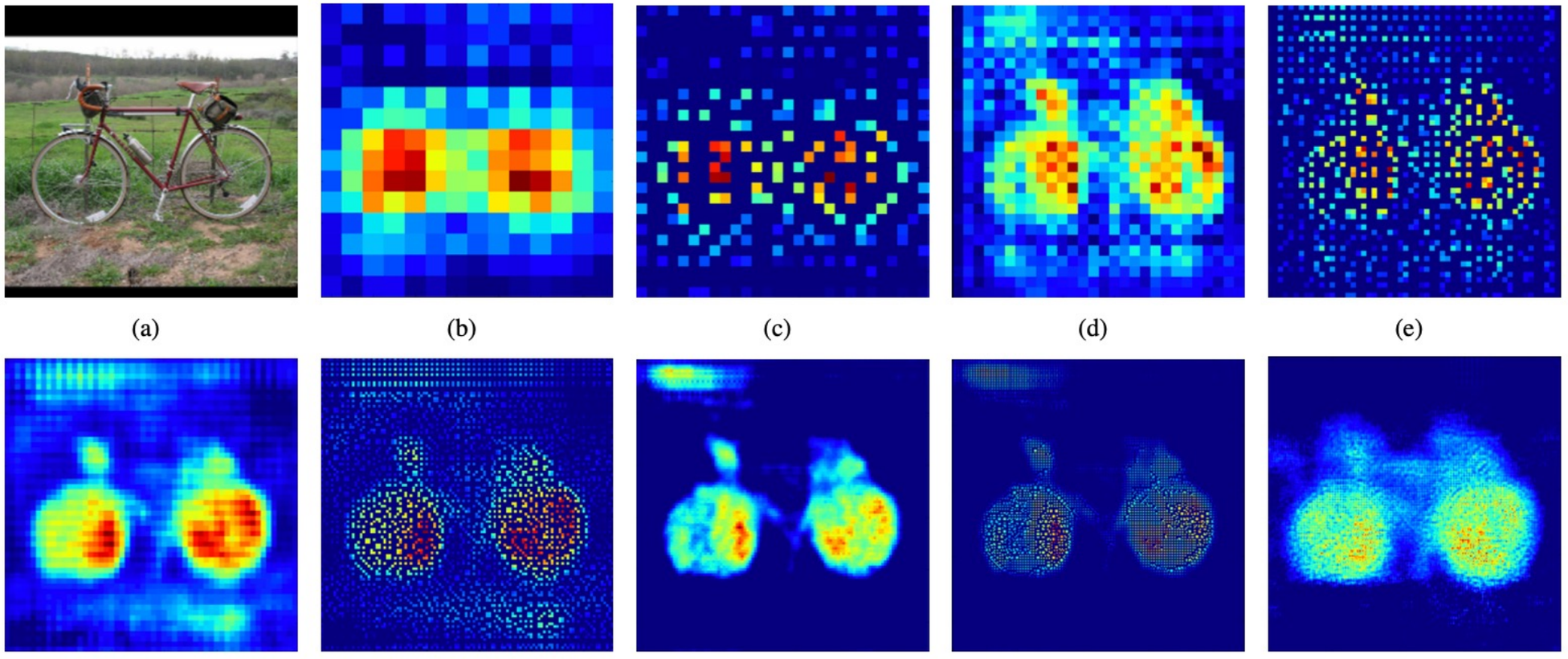
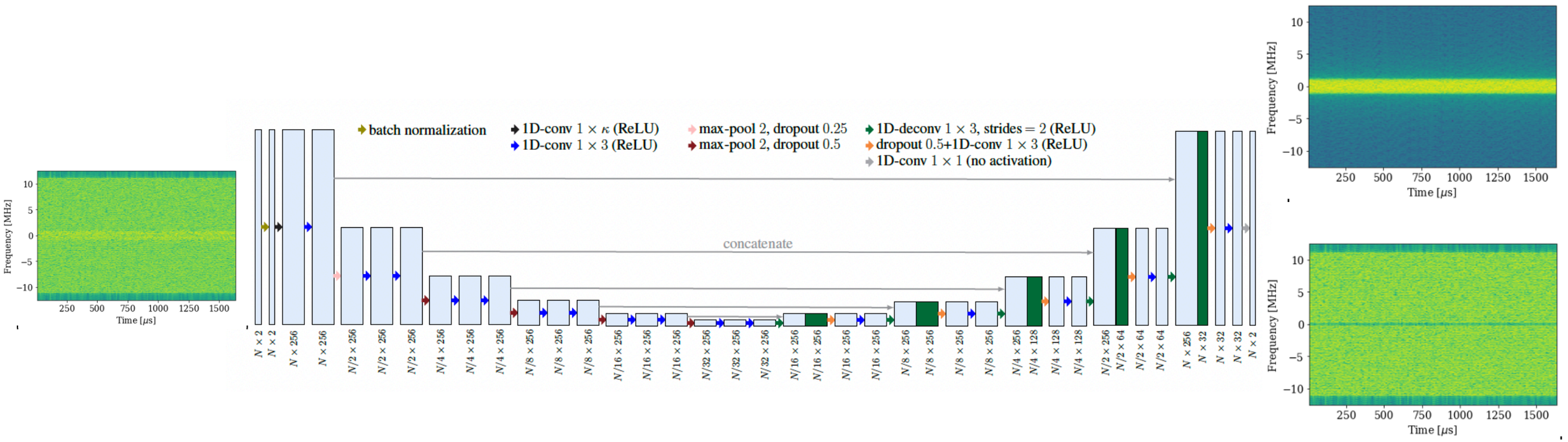


Figure 4. Visualization of activations in our deconvolution network. The activation maps from top left to bottom right correspond to the output maps from lower to higher layers in the deconvolution network. We select the most representative activation in each layer for effective visualization. The image in (a) is an input, and the rest are the outputs from (b) the last 14×14 deconvolutional layer, (c) the 28×28 unpooling layer, (d) the last 28×28 deconvolutional layer, (e) the 56×56 unpooling layer, (f) the last 56×56 deconvolutional layer, (g) the 112×112 unpooling layer, (h) the last 112×112 deconvolutional layer, (i) the 224×224 unpooling layer and (j) the last 224×224 deconvolutional layer. The finer details of the object are revealed, as the features are forward-propagated through the layers in the deconvolution network. Note that noisy activations from background are suppressed through propagation while the activations closely related to the target classes are amplified. It shows that the learned filters in higher deconvolutional layers tend to capture class-specific shape information.

"Learning Deconvolution Network for Semantic Segmentation", H. Noh, S. Hong and B. Han, ICCV, 2015

Applications: UNet for Source Separation of Radio Frequency Signals



- **Idea**: Use the **UNet autoencoder** structure as an **advanced denoiser**
- **Source separation** of two components equivalent to denoising
- Here data is quite different from images, but a **properly designed UNet can also work**

Video demo link here